

Exame de ultrassom FAST Scan-eFAST simplificado: Guia passo a passo

 pocus101.com/efast-ultrasound-exam-made-easy-step-by-step-guide

Vi Dinh

Autores principais: Seth White , [Vi Dinh](#) ; ***Coautores:*** Jessica Ahn, Jade Deschamps, Satchel Genobaga, Annalise Lang, Victor Lee, Devin Tooma , Reed Krause.
Supervisão, revisão e edição final por [Vi Dinh](#) (Editor de POCUS 101).

O exame **FAST** original , ou **Ultrassonografia Abdominal Focalizada para Trauma** (Rozycki, 1996), consistia em três incidências: o quadrante superior direito, o quadrante superior esquerdo e a pelve, para descartar sangramento abdominal decorrente de trauma. Também é comumente chamado de " **FAST Scan** " .

No entanto, em 2004, Kirkpatrick et al publicaram um artigo no Journal of Trauma que propôs a nomenclatura e o protocolo atuais de “**exame eFAST**” ou **Avaliação Focalizada Estendida com Ultrassonografia em Trauma** .

O exame eFAST incorpora a avaliação dos **pulmões** e **do coração** , além do **abdômen** . Isso ocorreu como resultado de diversos estudos publicados nas décadas de 1990 e 2000 que demonstraram a eficácia do uso do ultrassom para diagnosticar pneumotórax (Lichenstein et al), hemoperitônio (Kimura et al) e derrames pericárdicos (Plummer et al) no pronto-socorro.

Atualmente, o exame eFAST consiste em ultrassonografias para avaliar patologias no abdômen, pulmões e coração!

Neste post, mostraremos uma maneira fácil e eficiente de usar o ultrassom à beira do leito (POCUS) para:

- Realize o protocolo completo do exame de ultrassom eFAST em 5 etapas simples.
- Avaliar um paciente com suspeita de coleção de líquido livre intra-abdominal ou intratorácico
- Avaliar um paciente com suspeita de tamponamento cardíaco.
- Avaliar um paciente com suspeita de pneumotórax.

Índice

Indicações do exame de ultrassom eFAST

- Paciente traumatizado hemodinamicamente instável
- Trauma abdominal e torácico: contuso ou penetrante
- Paciente traumatizado previamente estável com piora aguda do quadro clínico.

Limitações do exame eFAST

- Não localiza o órgão abdominal lesionado.
- A visibilidade pode ser limitada em pacientes com enfisema subcutâneo.
- A visualização pode ser limitada em pacientes com lesão de víscera oca e presença de ar livre no abdômen.

Preparação para o exame de ultrassom eFAST

Preparação do Paciente

O paciente deve estar deitado **em decúbito dorsal** com a **mesa de exame plana** ou em **posição de Trendelenburg** (isso aumenta a sensibilidade do exame, mas *não é obrigatório*).



Aparelho de ultrassom para exame eFAST e posicionamento do paciente

Preparação da máquina

- **Transdutor** : O [transdutor phased array](#) (ou “sonda cardíaca”) pode ser usado durante *todo o exame*.
 - A técnica curvilínea também pode ser usada, *porém* , as imagens cardíacas podem ser difíceis de obter devido ao grande tamanho do transdutor.
 - Veja [aqui](#) para mais detalhes sobre a seleção de sondas.
- **Pré-configuração** : exame FAST ou exame abdominal.
- **Posicionamento do aparelho de ultrassom**: Posicione o aparelho no lado direito do paciente. Isso permite que você faça o exame com a mão direita e opere os controles do ultrassom com a mão esquerda.

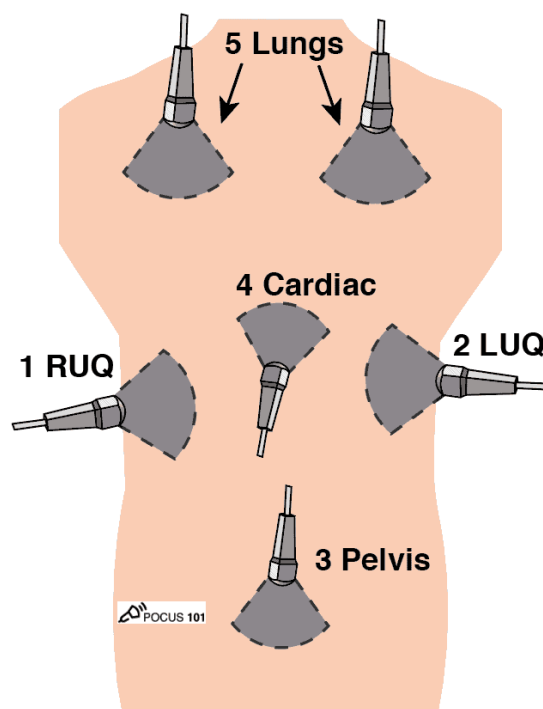
Se você precisar de uma breve revisão sobre como manipular a sonda de ultrassom antes de começarmos, [clique AQUI](#) !

Sequência recomendada para o exame eFAST

Existem muitas sequências e variações diferentes descritas na literatura. Escolha uma sequência que funcione para você e certifique-se de executá-la sempre da mesma maneira para não se esquecer de nenhuma das vistas.

Esta é a ordem de visualização do exame eFAST que realizamos em nossa instituição e que recomendamos:

1. **Vista do quadrante superior direito (QSD)**
2. **Vista do quadrante superior esquerdo (QSE)**
3. **Vista pélvica**
4. **Vista cardíaca** (eixo longo paraesternal ou subxifoide)
5. **Pulmões** (Direito e Esquerdo)



Sequência recomendada para o exame eFAST

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Protocolo passo a passo do exame eFAST

O exame eFAST serve para responder a 4 perguntas:

1. Does my patient have free fluid in the **Abdomen**?
2. Does my patient have free fluid in the **Thorax**?
3. Does my patient have fluid in the **Pericardium**?
4. Does my patient have a **Pneumothorax**?

We will first introduce you to the 5 steps of how to perform the eFAST ultrasound exam and then in the following section we will give you examples of all of the major eFAST pathologies you may encounter.

Step 1: eFAST Right Upper Quadrant View (RUQ)

For the right upper quadrant and left upper quadrant views. Remember to look free fluid both above (hemothorax) and below (hemoperitoneum) the diaphragm.

| eFAST RUQ View: Does my patient have free fluid in the abdomen or right thorax?

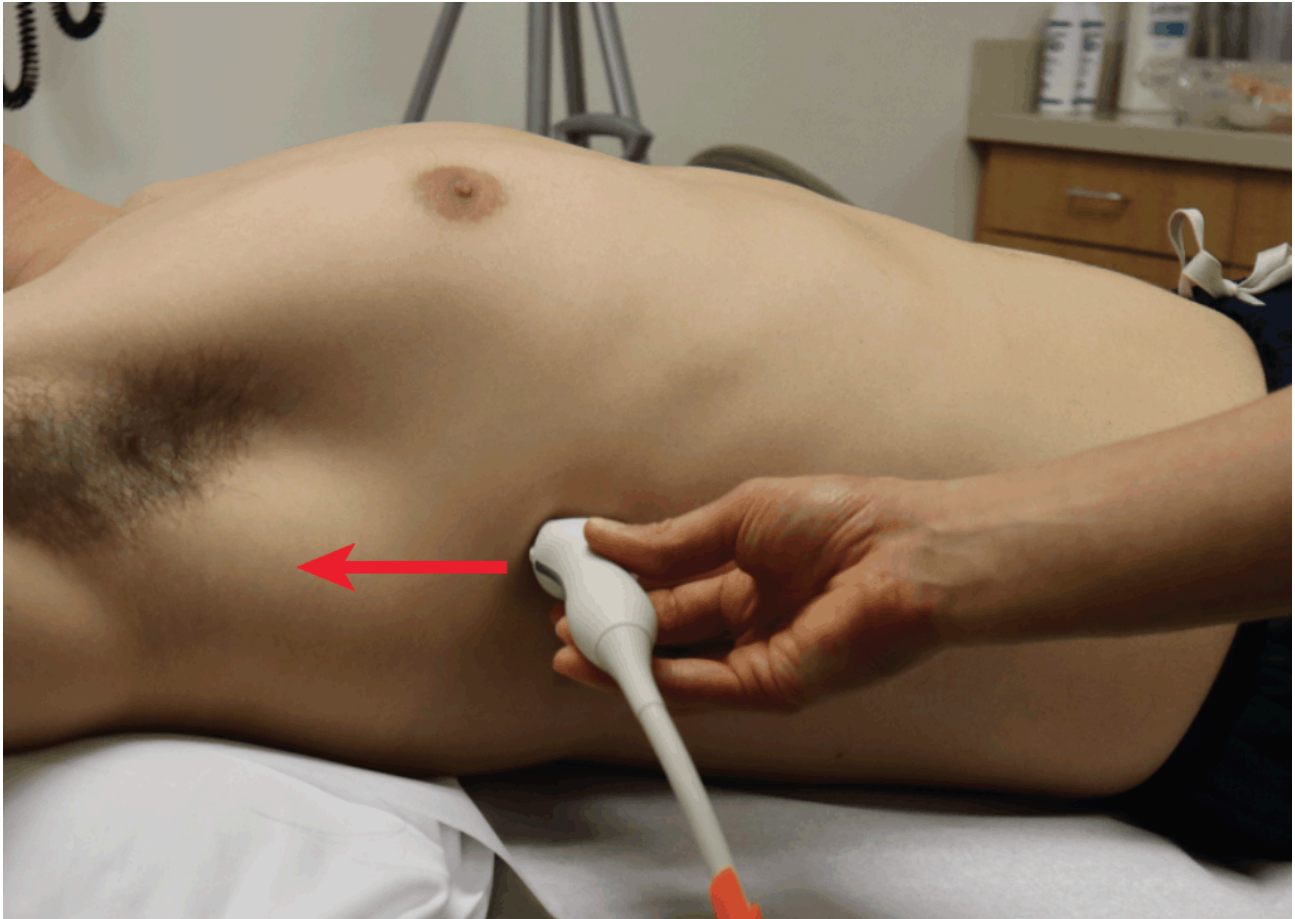
Since the liver is the most commonly injured organ in blunt abdominal trauma, the right upper quadrant is usually the most sensitive view of the eFAST exam.

RUQ Probe Position and Hand Placement

- Orientate the **probe indicator** towards the **patient's head**.
- **Anchor** your probe in the **midaxillary line** at the **10th intercostal space**.



eFAST Exam Probe Placement – RUQ

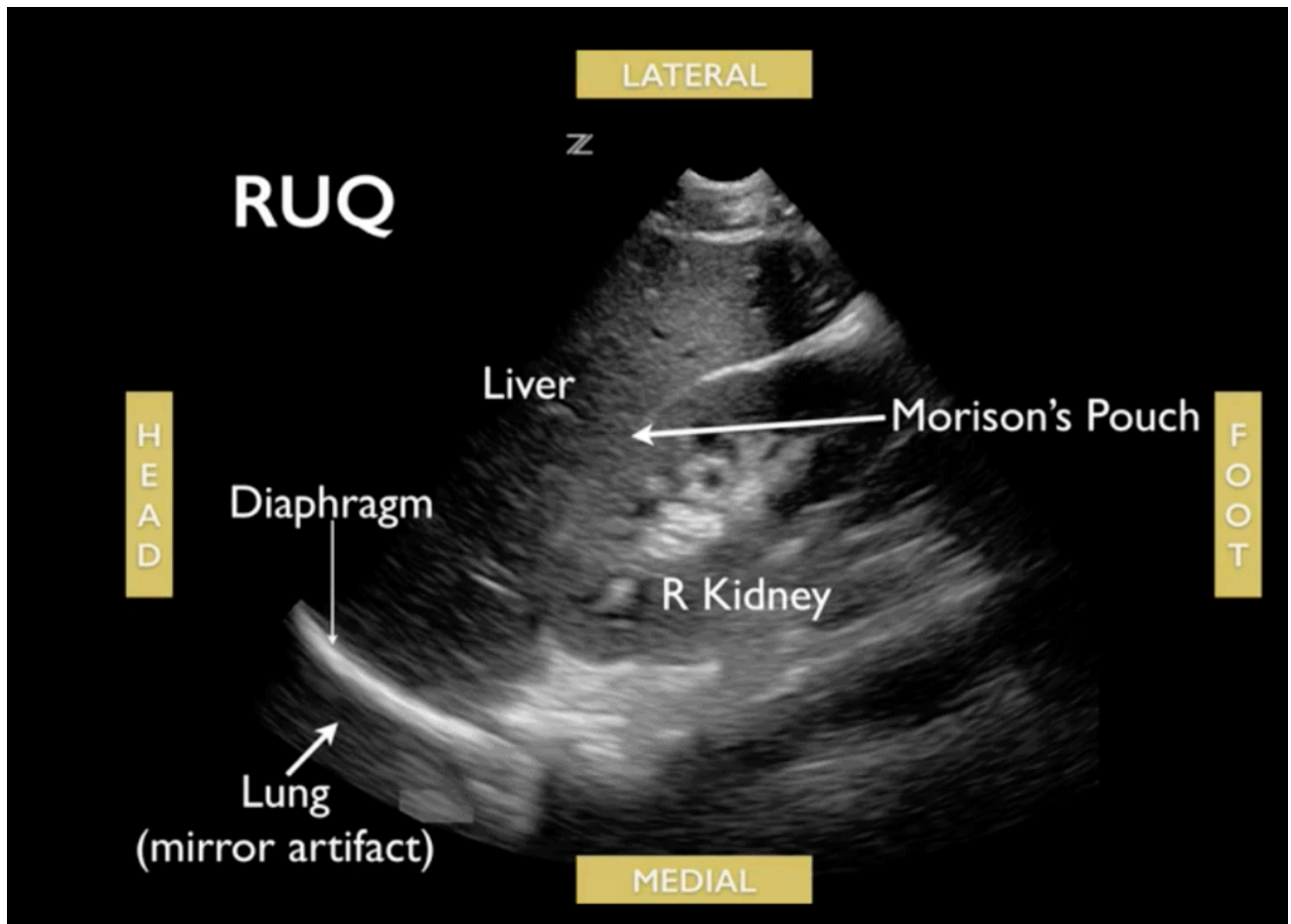


eFAST Exam Probe Placement – RUQ

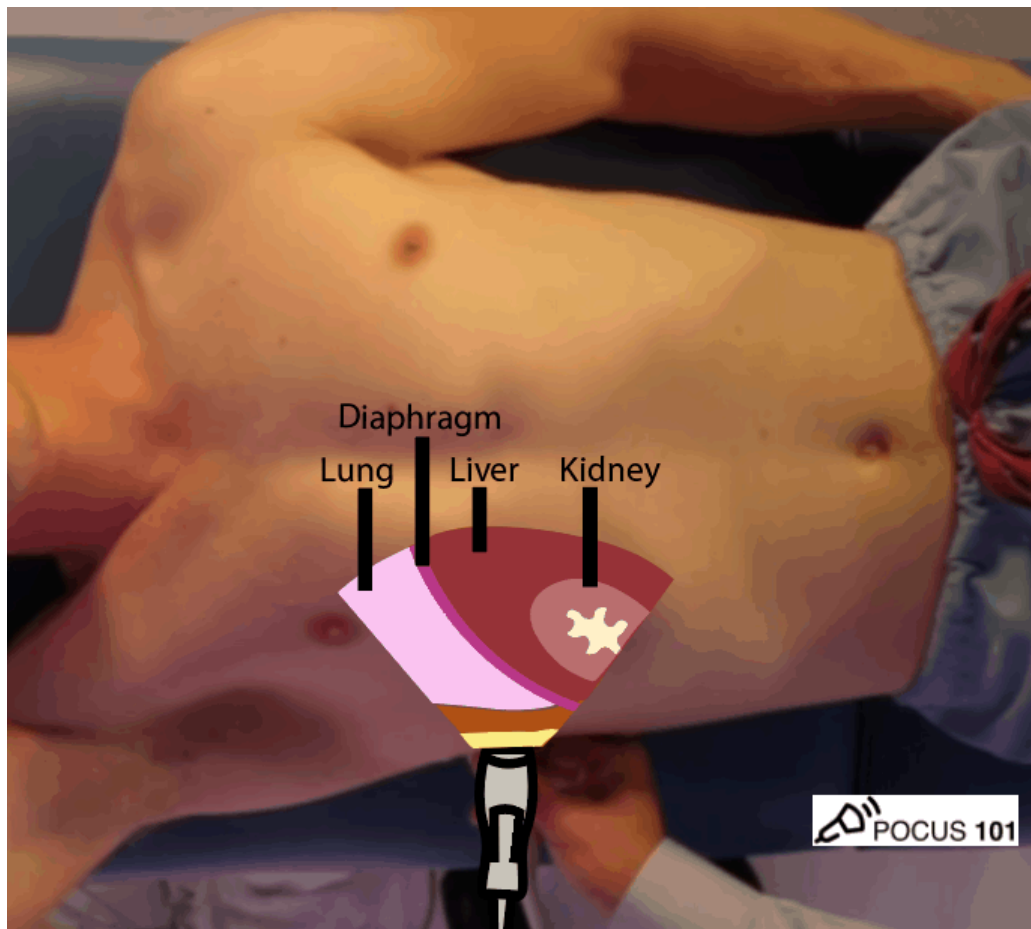
RUQ Normal View and Structures

- Using the *liver as an acoustic window*, identify the **lung, liver, Morison's Pouch, diaphragm**, and the long-axis of the **right kidney**.
- **Morison's Pouch** is where you usually identify free fluid in the RUQ view.
- A Mirror Image Artifact is a normal finding signifying there is an aerated lung above the diaphragm. If you want to dive a little deeper into the physics of the Mirror Image Artifact, [click here](#).
- You may have to **slide up** or **down** a rib space to identify the structures.

POCUS 101 Tip: these structures move as the diaphragm contracts and relaxes during the respiratory cycle. Consider asking your patient to hold their breath to keep the desired organs from moving. Also, consider slightly rotating the probe counterclockwise towards the bed so that the probe fits better between the rib spaces.



Normal RUQ – eFAST exam



RUQ eFAST exam illustration

Step 2: eFAST Left Upper Quadrant View (LUQ)

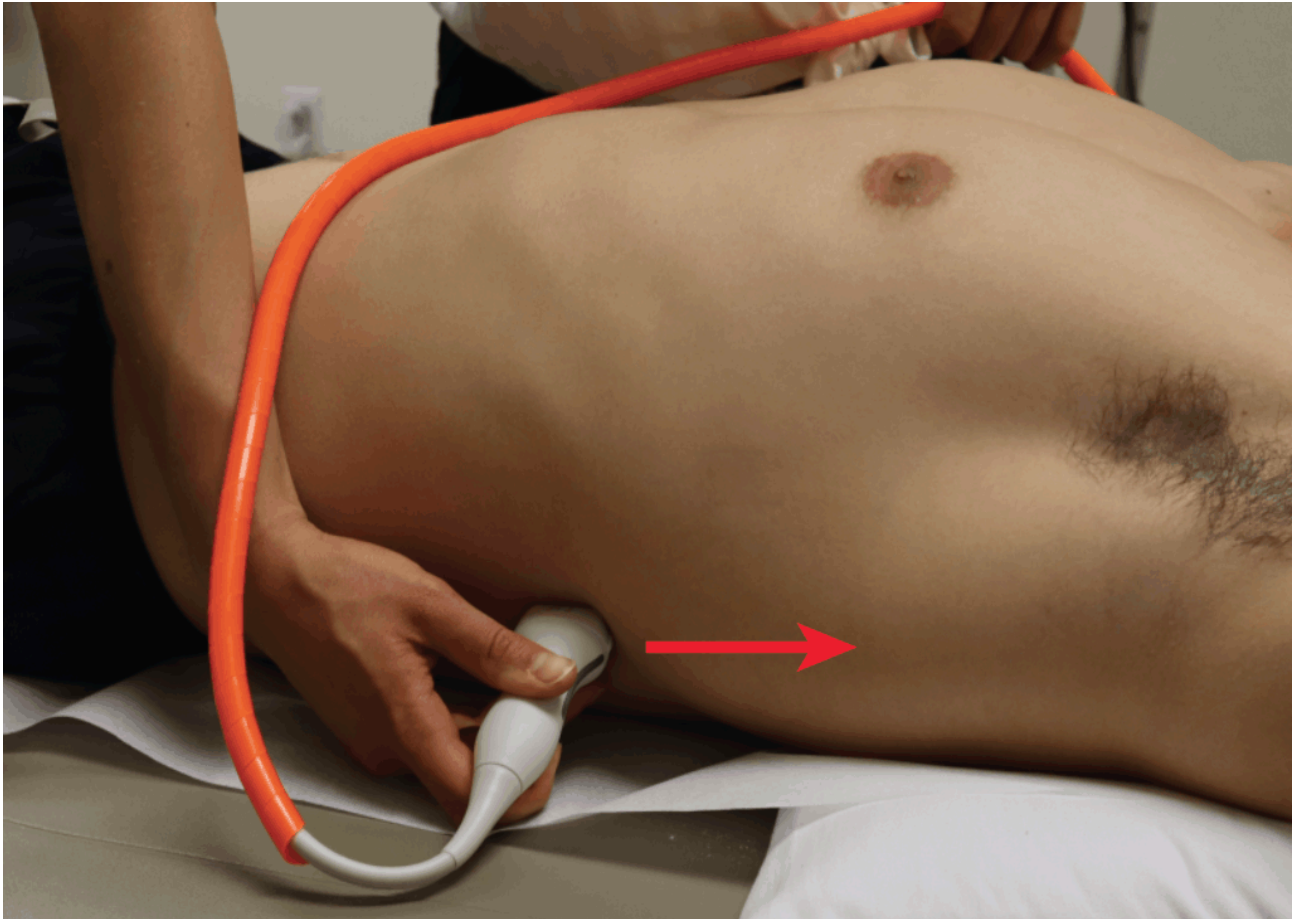
eFAST LUQ View: Does my patient have free fluid in the abdomen or left thorax?

LUQ Probe Position and Hand Placement

- Grasp the probe between your **thumb** and **first finger**, *like holding a pencil*.
- Orientate the **probe indicator** towards the **patient's head**.
- Anchor your probe in the **posterior axillary line** around the **8th intercostal space**.
- You should have your “*Knuckles to the bed*” since the spleen is fairly posterior.



eFAST Exam Probe Placement – RUQ

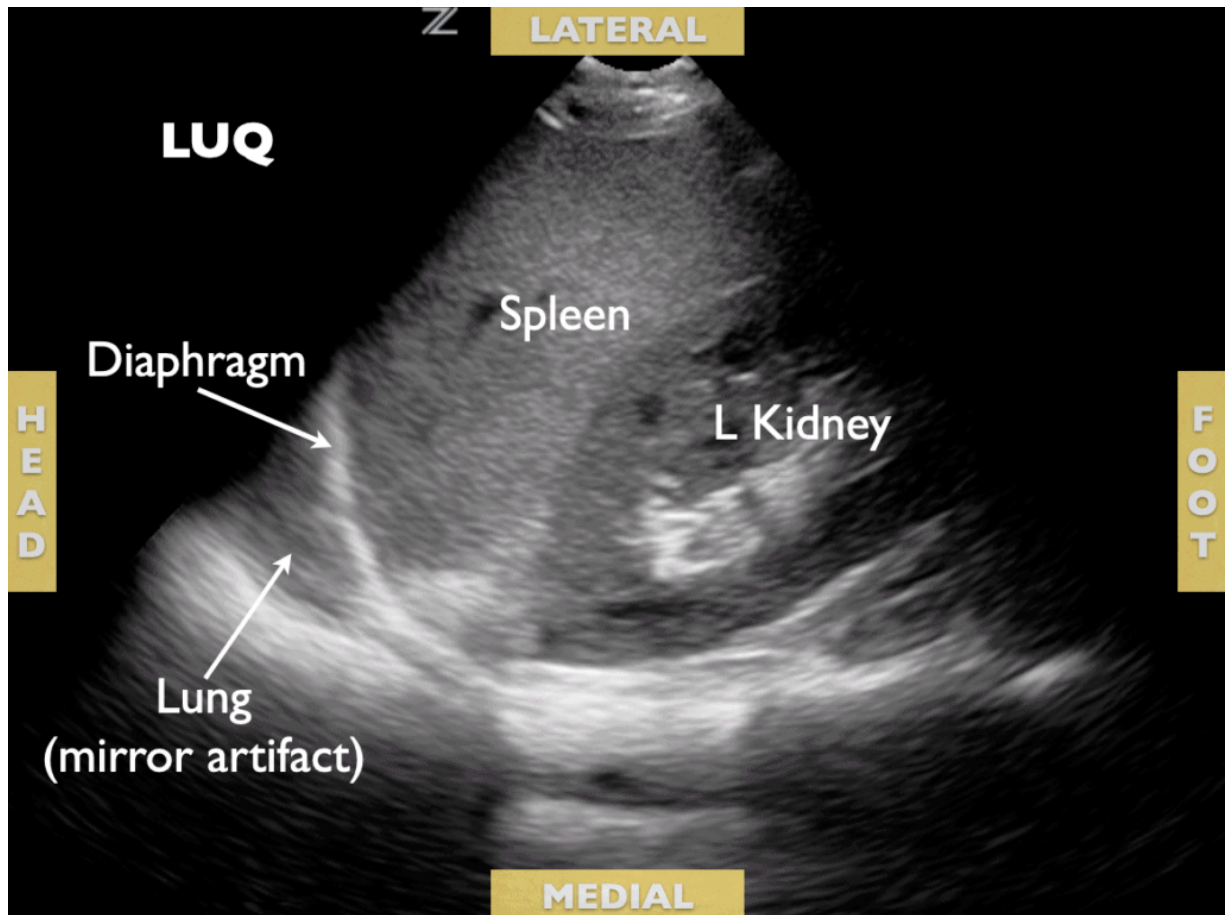


“Knuckles to the Bed” for the LUQ eFAST exam view

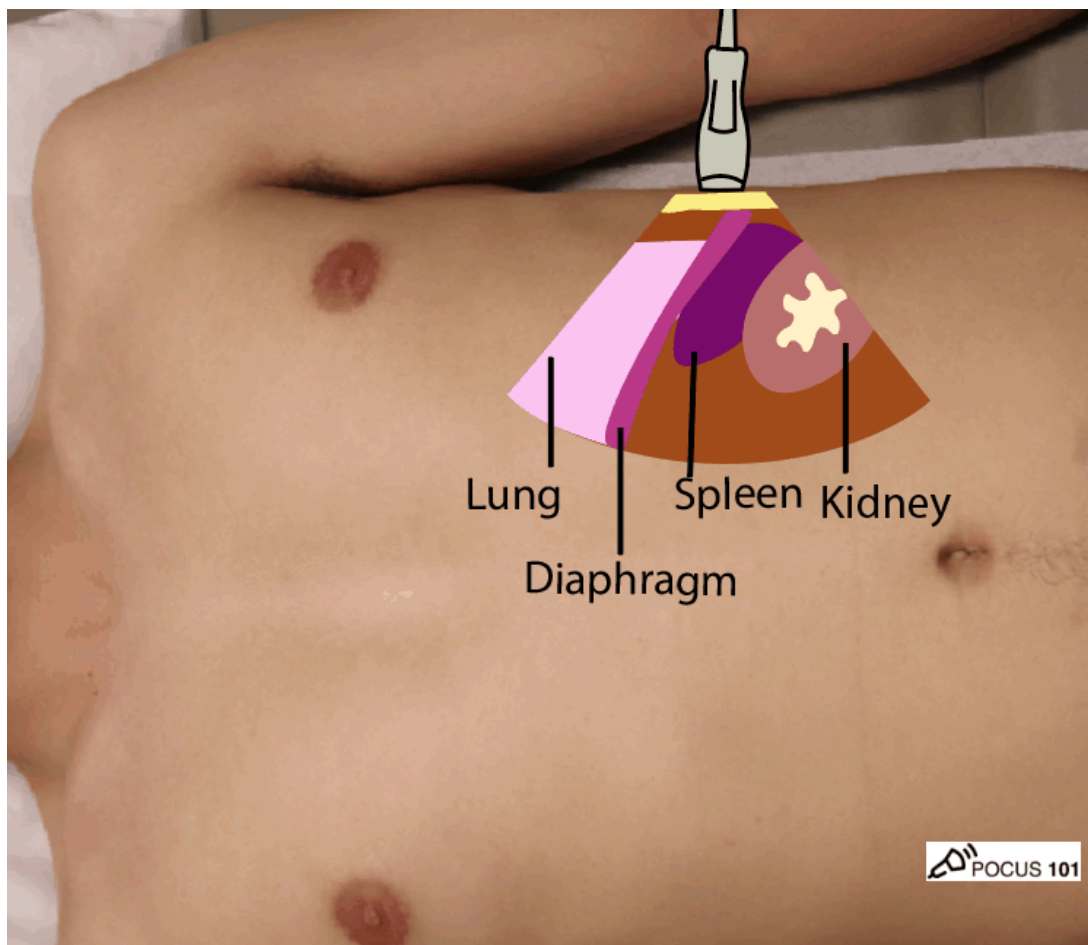
LUQ Normal View and Structures

- Using the *spleen as an acoustic window*, identify the **spleen**, **perisplenic space**, **diaphragm**, and the long-axis view of the **left kidney**.
- Free fluid in the LUQ is most frequently seen in the perisplenic space (between the spleen and the diaphragm). The reason is that there is a splenorenal ligament limiting the ability of fluid to track in between the spleen and left kidney
- A **Mirror Image Artifact** is a normal finding, similar to the RUQ, signifying there is an aerated lung above the diaphragm. If you want to dive a little deeper on this, [see here](#).
- You may have to **move up** or **down** a rib space to identify the structures.

POCUS 101 Tip: these structures move as the diaphragm contracts and relaxes during the respiratory cycle. Consider asking your patient to hold their breath to keep the desired organs from moving. Also, consider slightly rotating the probe clockwise towards the bed, so that the probe fits better between the rib spaces.



Normal LUQ – eFAST exam



LUQ eFAST exam illustration

Step 3: eFAST Pelvic View

| eFAST Pelvic View: Does my patient have free fluid in the abdomen or pelvis?

When looking for free fluid in the pelvis it's important to consider the sex of your patient, as free fluid has a tendency to accumulate in different locations depending on the patient's gender. Additionally, females may have a scant amount of normal physiologic fluid accumulation in the pelvis (pouch of Douglas).

- Ideally, scan your patient with a **full bladder** (so the bladder can be used as an acoustic window). Therefore, try to initiate the scan **prior to foley catheter placement**.
- Machine **Depth: 10-15 cm**

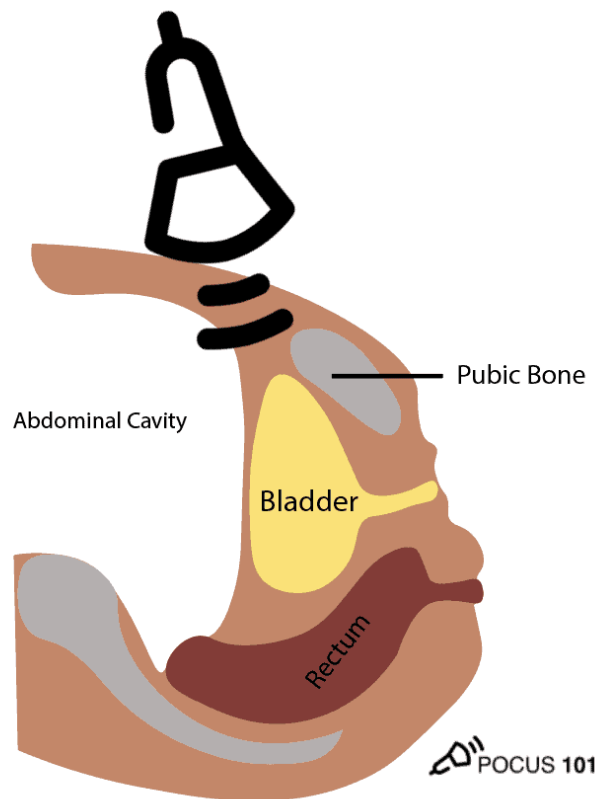
Pelvic Ultrasound – Longitudinal View

- Place the transducer with the **indicator pointing towards the patient's head** in the patient's midline, right **above the pubic symphysis**.
- Rock the probe so that it **points down towards the pelvic cavity**.



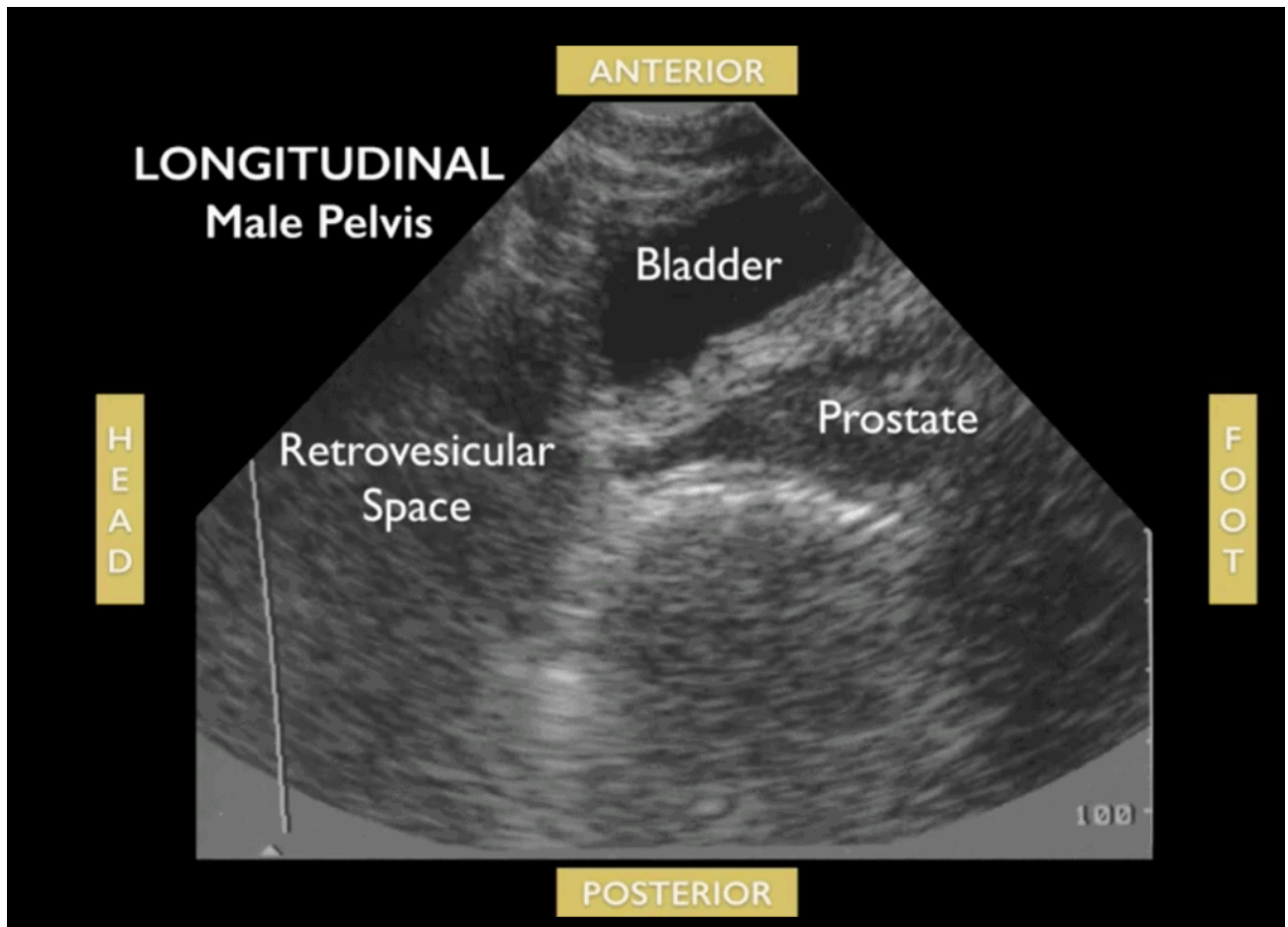
eFAST Pelvic Ultrasound – Longitudinal View

POCUS 101 Tip: One of the most important things to remember when performing pelvic ultrasound in the eFAST scan is that the bladder is directly posterior to the pubic bone/symphysis. If you are unable to get proper images, most likely your ultrasound probe is placed too superiorly.

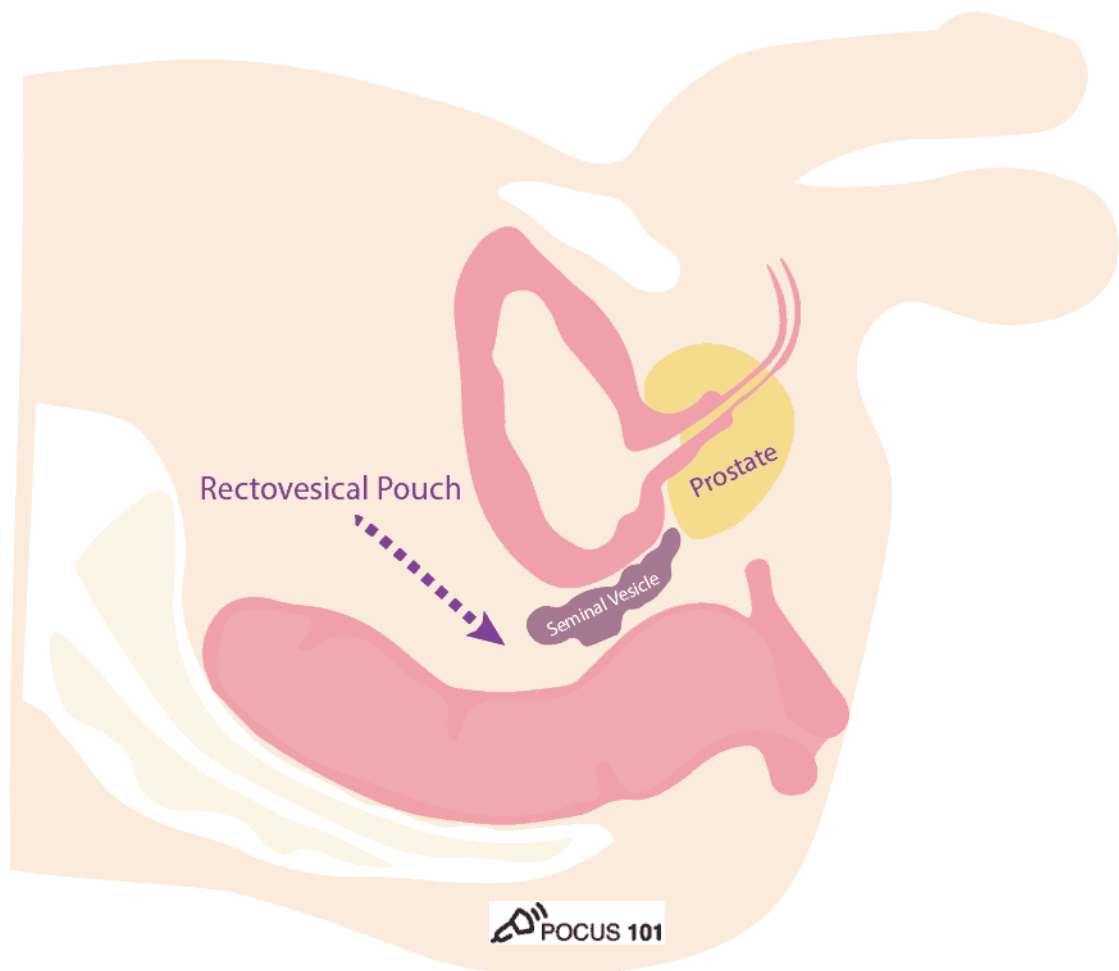


Make sure to point your probe into the pelvis!

- In males, identify the **bladder** (*immediately posterior to the symphysis*), **prostate/seminal vesicle**, and **rectovesical pouch** in the longitudinal view.
- The rectovesical pouch is where free fluid will accumulate in the male pelvis.

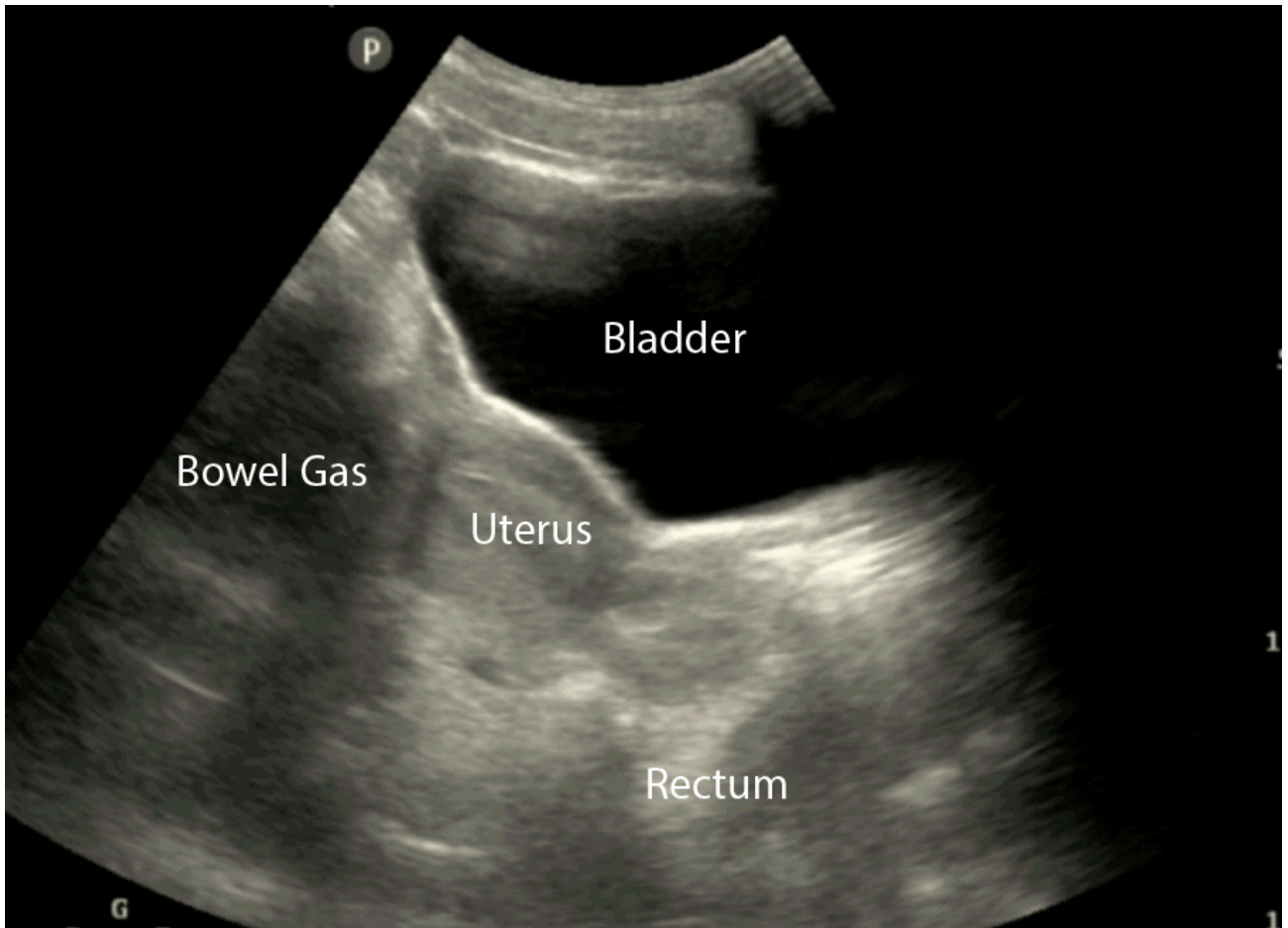


Male eFAST exam Pelvic View – Longitudinal

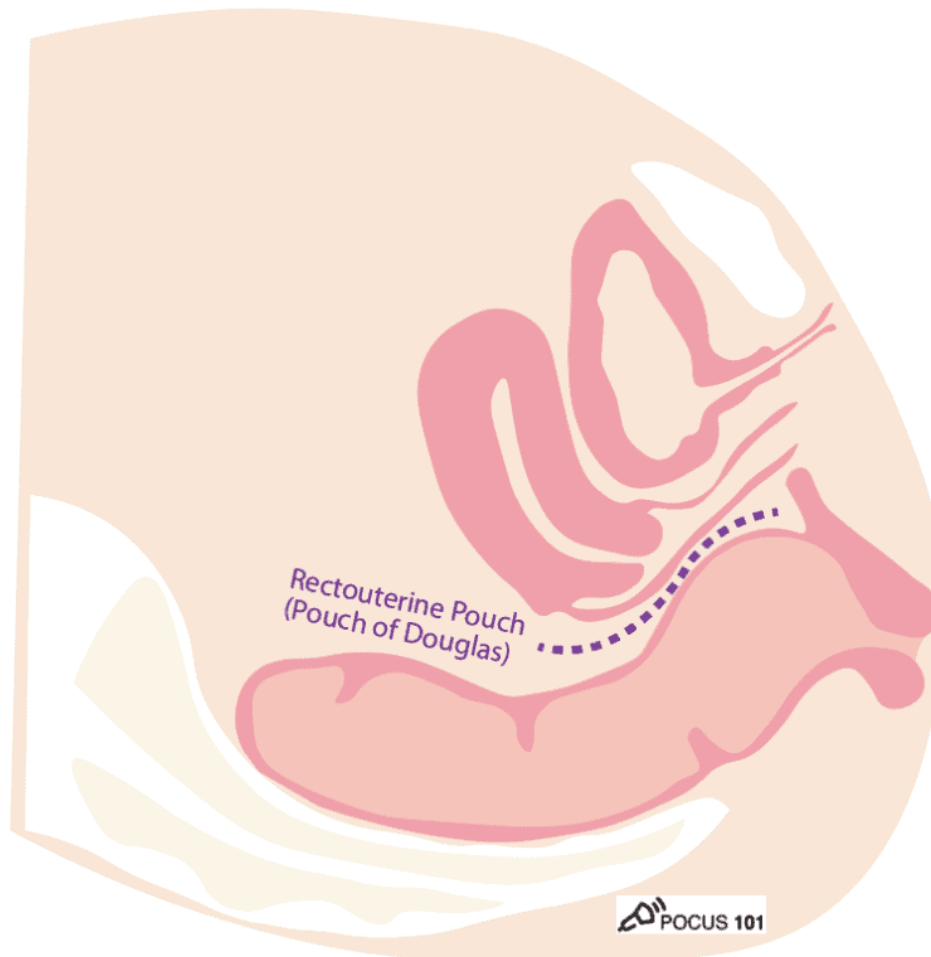


Male eFAST exam Pelvic View – Longitudinal (Illustration)

- In Females, identify the **bladder**, **uterus**, and **Rectouterine Pouch** (also called the *Pouch of Douglas*).
- The Pouch of Douglas is where free fluid will accumulate in the female pelvis.



Female eFAST exam Pelvic View – Longitudinal



Female eFAST exam Pelvic View – Longitudinal (Illustration)

In all patients (male or female), observe the lateral borders of the bladder to identify free fluid by [tilting/fanning](#) the probe left and right.



Tilting the probe in the longitudinal view

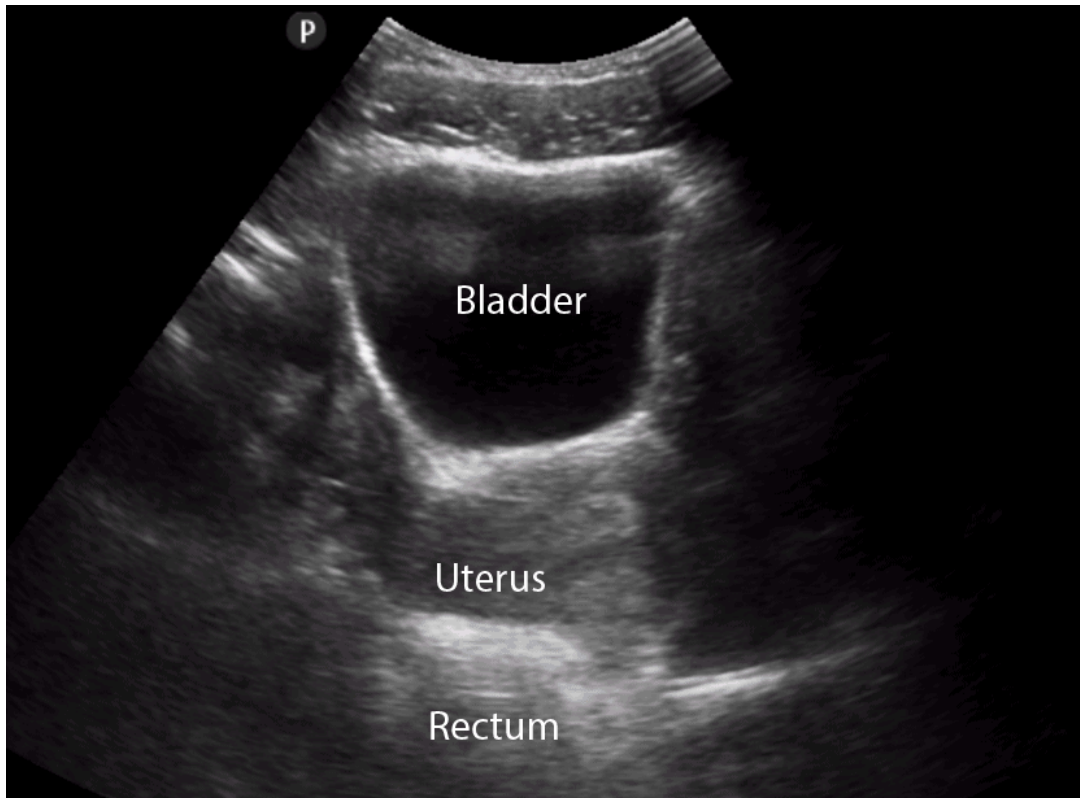
Pelvic Ultrasound – Transverse View

- Next, center the bladder and then rotate the transducer **90 degrees counterclockwise**. The indicator should now point to the patient's **Right side**.
- Make sure to tilt the ultrasound probe so it scans into the pelvic cavity.

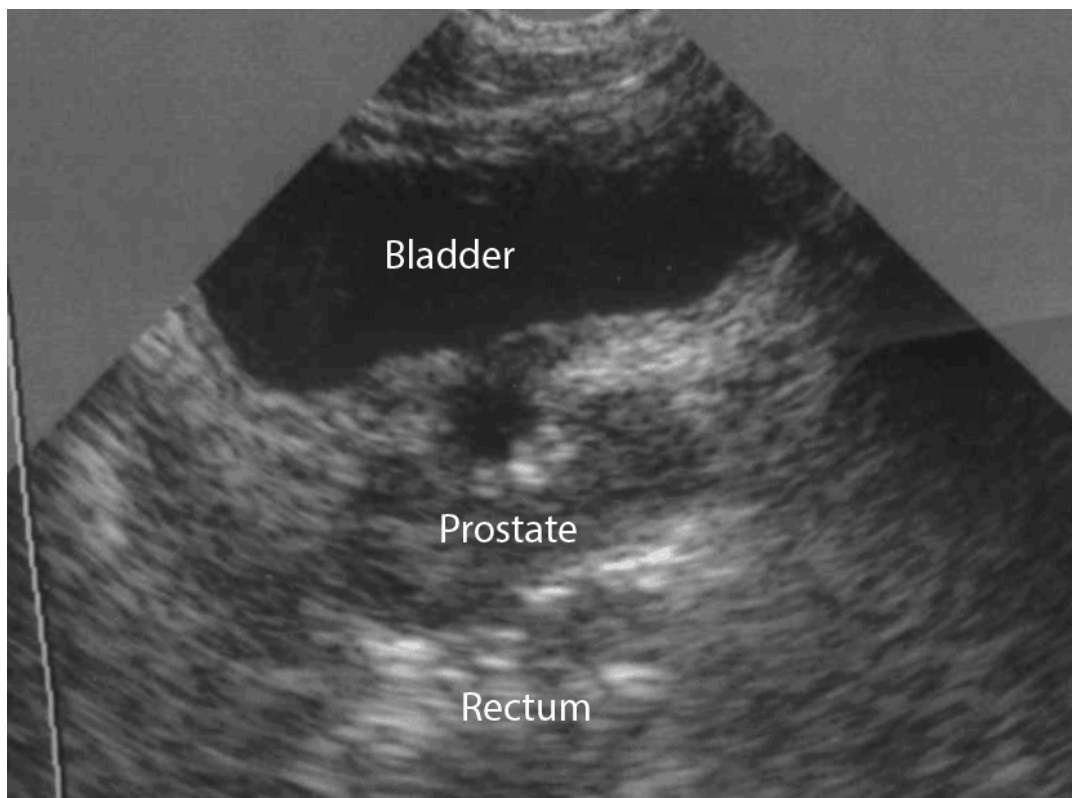


eFAST Pelvic Ultrasound – Transverse View

In the [transverse](#) view, identify the **Bladder**, **Uterus** (females), **Prostate** (males), and **Rectum**.

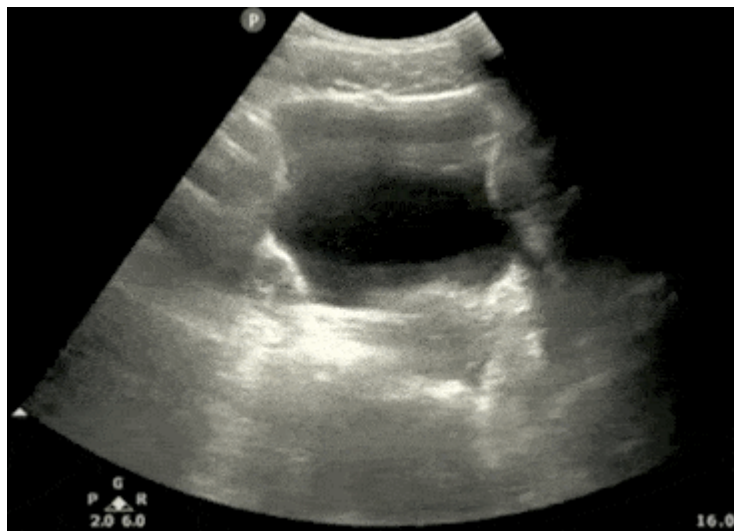


Female eFAST exam Pelvic View – Transverse



Male eFAST exam Pelvic View – Transverse

Tilt/Fan the probe to examine the entire pelvis from superior to inferior.



Transverse view of the eFAST Pelvic View

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Step 4: eFAST Cardiac View

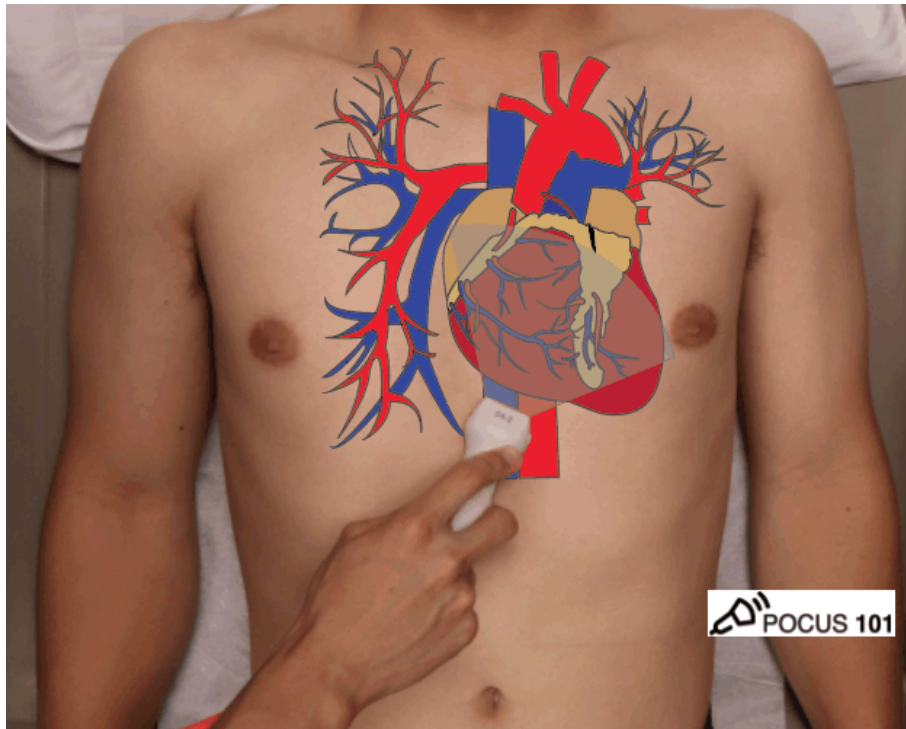
eFAST Cardiac View: Does my patient have a pericardial effusion with cardiac tamponade?

The subxiphoid view is the preferred scan to evaluate the heart and assess for any pericardial fluid. However, the subxiphoid view can be difficult to obtain in some situations (e.g. patient body habitus or abdominal pain with trauma). If the subxiphoid view cannot be obtained, a parasternal long axis view is recommended.

**Editor's Note: In this post, we will be pointing the indicator based on the abdominal/eFAST ultrasound preset. When you are in the dedicated cardiac mode preset the indicator will be pointing 180 degrees in the opposite direction.*

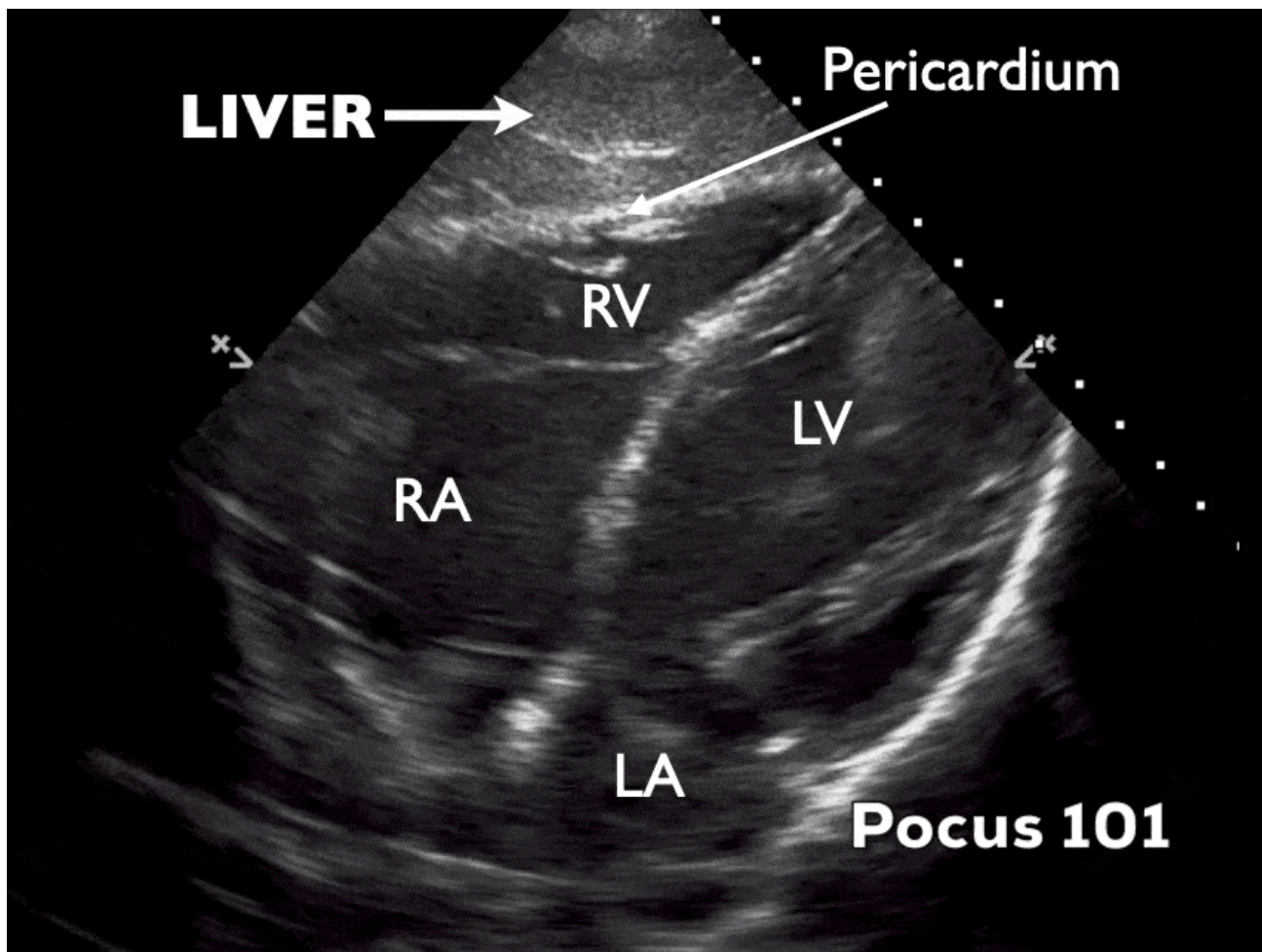
Cardiac Subxiphoid View

- Hold the probe in the palm of your hand and use an overhand grip.
- **Point the probe indicator** towards the **patient's right** with the ultrasound machine **depth** set to around **15-20 cm**.
- Using the *liver as the acoustic window*, **simultaneously** press the probe into the patient's abdomen while tilting the **tail of the probe towards the patient's feet**
- **Aim the ultrasound beam towards the patient's left shoulder.**

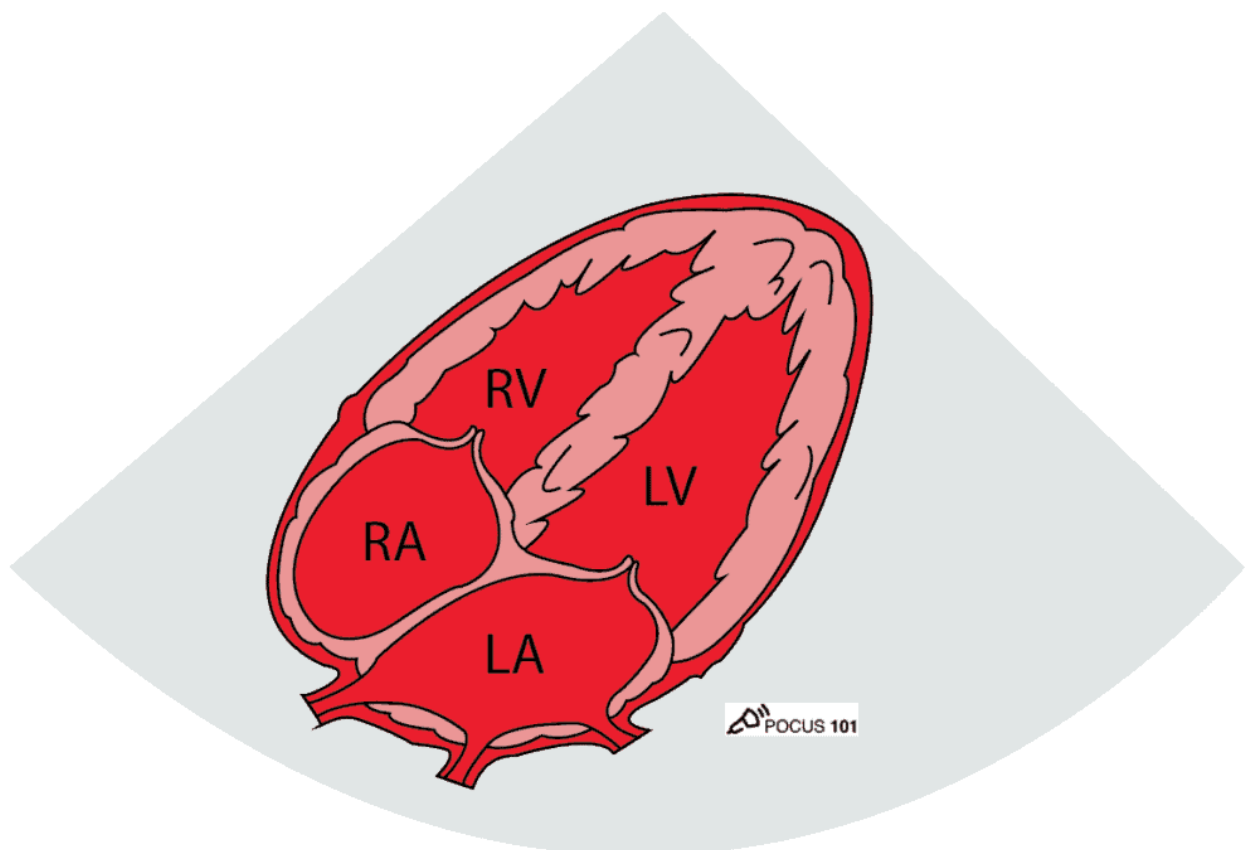


eFAST Subxiphoid View Probe Position

Identify the **liver**, **pericardium**, **right atrium**, **right ventricle**, **left atrium** and **left ventricle**



eFAST Subxiphoid View Structures

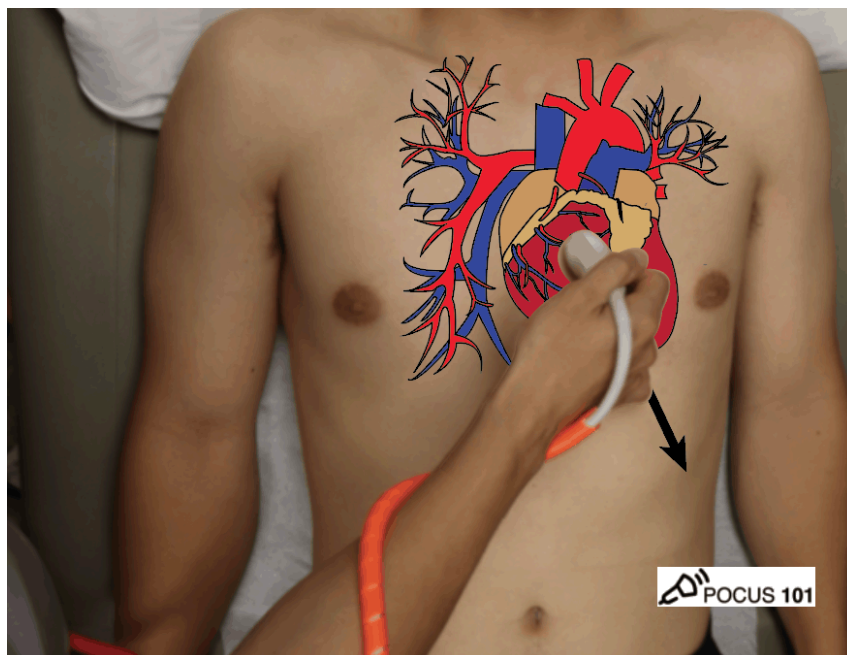


eFAST Subxiphoid View Structures (Illustration)

Cardiac Parasternal Long Axis

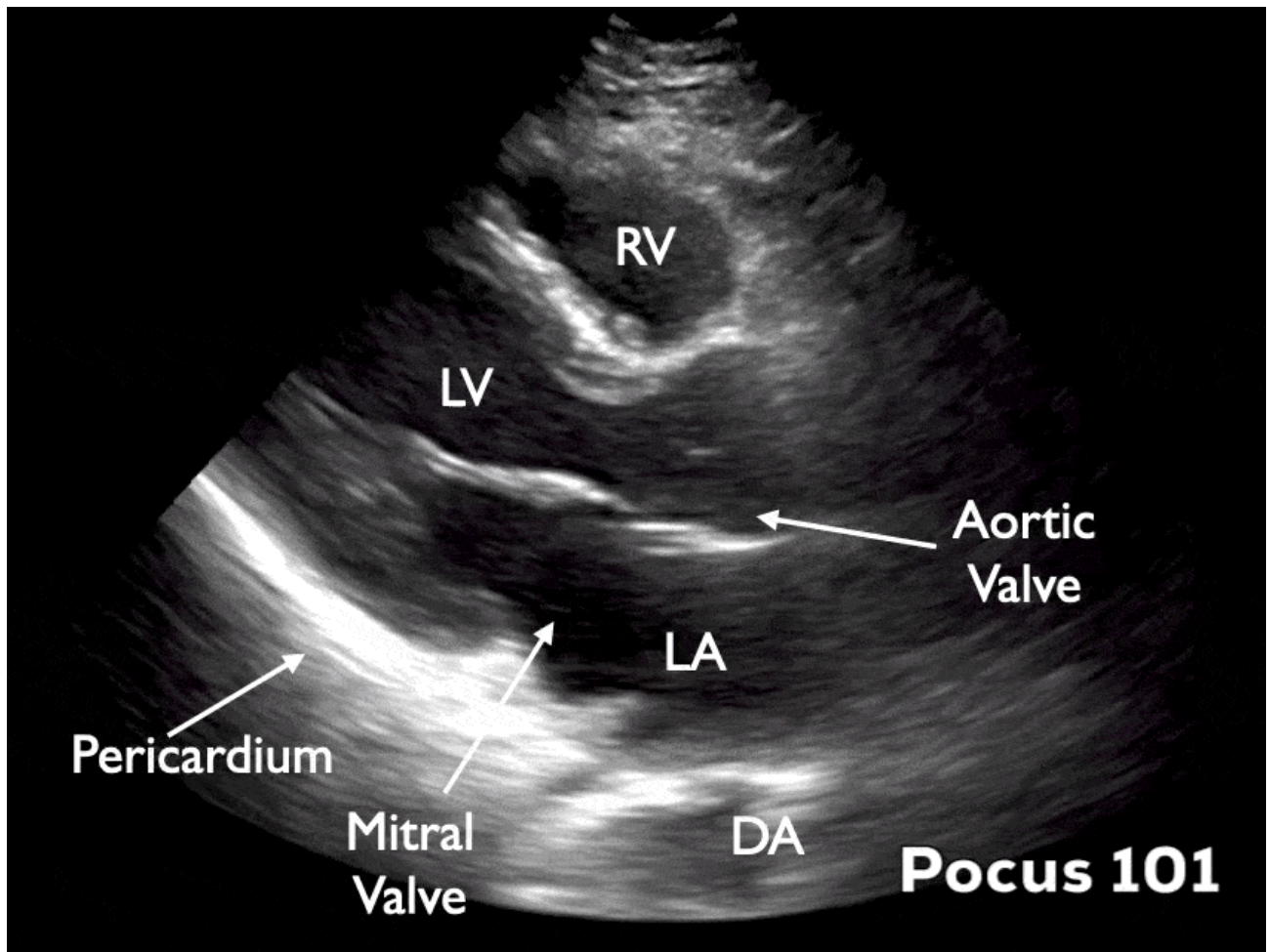
Consider obtaining the parasternal long-axis view of the heart in a patient when you are unable to obtain the subxiphoid view. Otherwise, proceed to scanning the lungs.

- Grasp the linear probe between your **thumb** and **first finger**, like *holding a pencil*.
- **Anchor** your third and/or fourth finger(s) in the **2nd or 3rd left intercostal space**, just **lateral** to the **sternum**.
- **Probe indicator** towards the patient's **left hip** with the machine **depth** set approximately **10-15 cm**

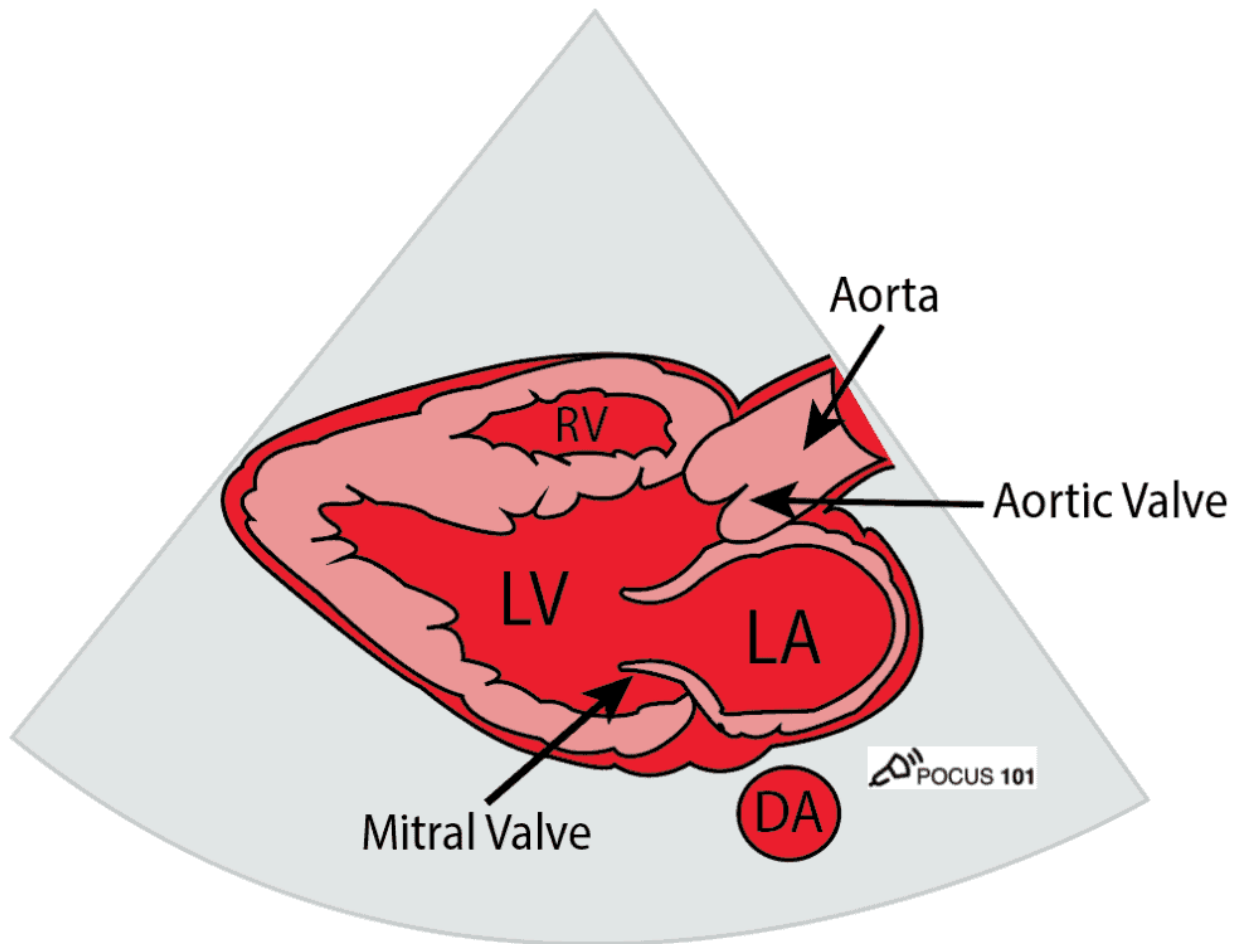


eFAST Parasternal Long Axis View Probe Position

- Identify the **pericardium, mitral valve, aortic valve, right ventricle, descending aorta, left atrium, and left ventricle**
- You may have to **move up** or **down** a rib space to identify the structures



eFAST Parasternal Long Axis View Structures



eFAST Parasternal Long Axis View Structures (Illustration)

Step 5: eFAST Lung Views

| eFAST Lung Views: Does my patient have a pneumothorax?

Recall, the **parietal pleura** covers the surface of the thorax and the **visceral pleura** covers the lungs. The presence of these moving against one another produces **lung sliding** on ultrasound. For a full tutorial on lung ultrasound [click HERE](#).

Lung Probe Position and Hand Placement

- Point your indicator towards the patient's head.
- Place your probe at the **mid-clavicular line at the 2nd intercostal space of the right and left lungs respectively**. This point is the most sensitive spot for looking for pneumothorax in the supine patient.
- Anchor your probe in the space between two ribs and set the ultrasound machine depth to 3-5cm.



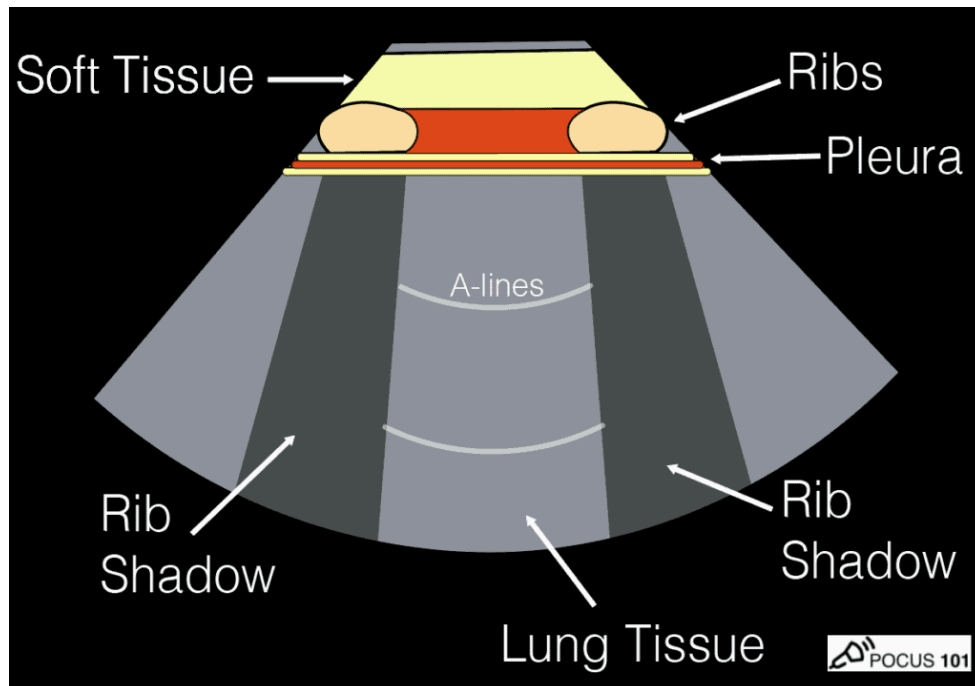
eFAST Scan of the Right Lung



eFAST Scan of the Left Lung

Identify Two Rib Shadows (Batwing Sign)

The first lung ultrasound finding to confirm you are in the correct position is to look for the two rib shadows or the “**Batwing Sign**.” This ensures that your probe is in between two ribs.



Batwing Sign Illustration



Batwing Sign on Ultrasound

Identify Lung Sliding

- The next finding you will want to look for is **lung sliding** during respiration.

- Lung sliding is a normal finding where the visceral and parietal pleura slide back and forth on one another as the patient breathes. Some say this looks like tiny “**ants marching on a line.**”
- “**Ants Marching**” sign is produced from the visceral and parietal pleura moving against one another during respiration.
- This is a simple finding but extremely useful since lung sliding definitely means that the visceral and parietal pleura are next to each other, effectively ruling out a pneumothorax.

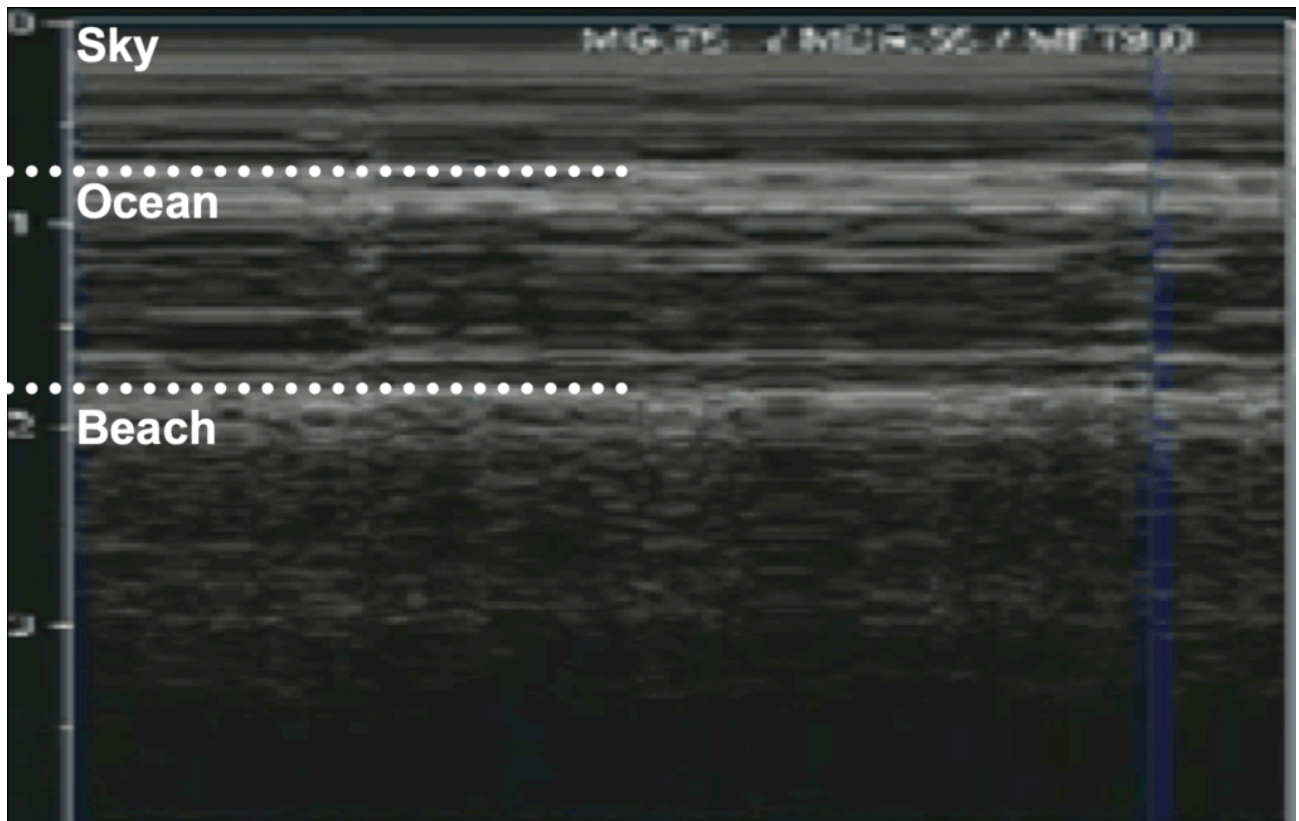


Lung Sliding

Although looking for lung sliding using B-Mode is usually sufficient, you can also consider assessing for a pneumothorax using M-mode.

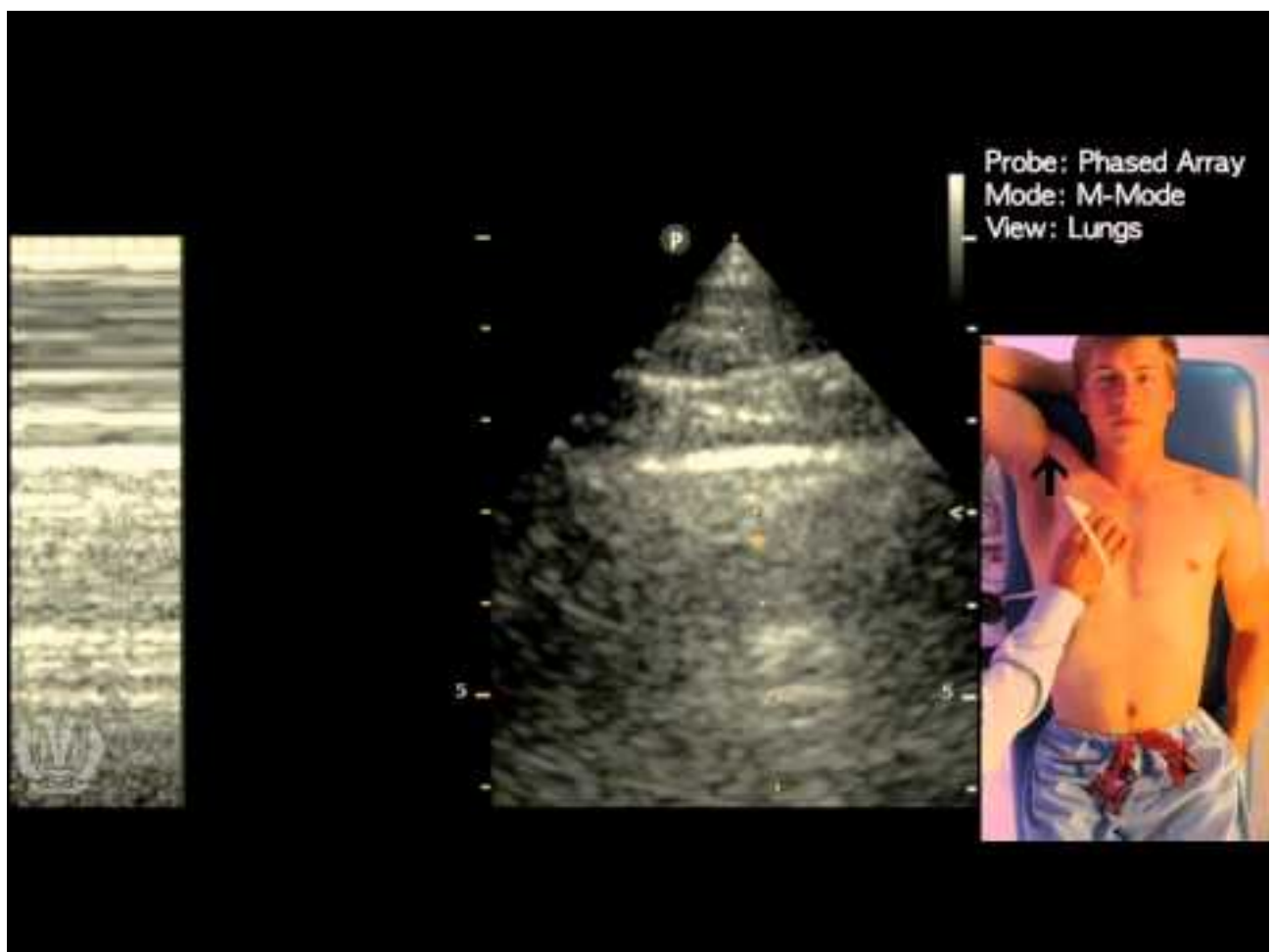
Identify Lung Sliding using M-Mode (Optional)

- If lung sliding is not readily apparent, it can be further be evaluated using [M-Mode](#). The goal of M-Mode is to see if the patient has a **normal seashore sign**.
- Place the ultrasound machine in **M-Mode** (*M-Mode for motion*)
- Place the ultrasound doppler indicator/cursor over the **lung field** (**NOT over the rib**).
- Look for the normal “**Seashore Sign**”: Sky = Skin/Subcutaneous Tissue, Ocean= Muscle, Beach = Lung sliding motion (sandy appearance). The Seashore Sign is a NORMAL finding.
- Contrast this with the [Barcode sign](#) (see in eFAST pathology section below).



Normal Lung with Seashore Sign on Lung Ultrasound.

Here is a quick video on how to use M-mode for lung ultrasound:



[Watch on YouTube](#)

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eFAST Ultrasound Pathology

Recall that fluid will appear black, or **anechoic**. For the purposes of the eFAST exam we are looking for anechoic (black) areas in the abdomen, chest, and heart that signify bleeding in those potential spaces.

For pneumothorax we will be evaluating the presence or absence of lung sliding.

Hemoperitoneum – eFAST

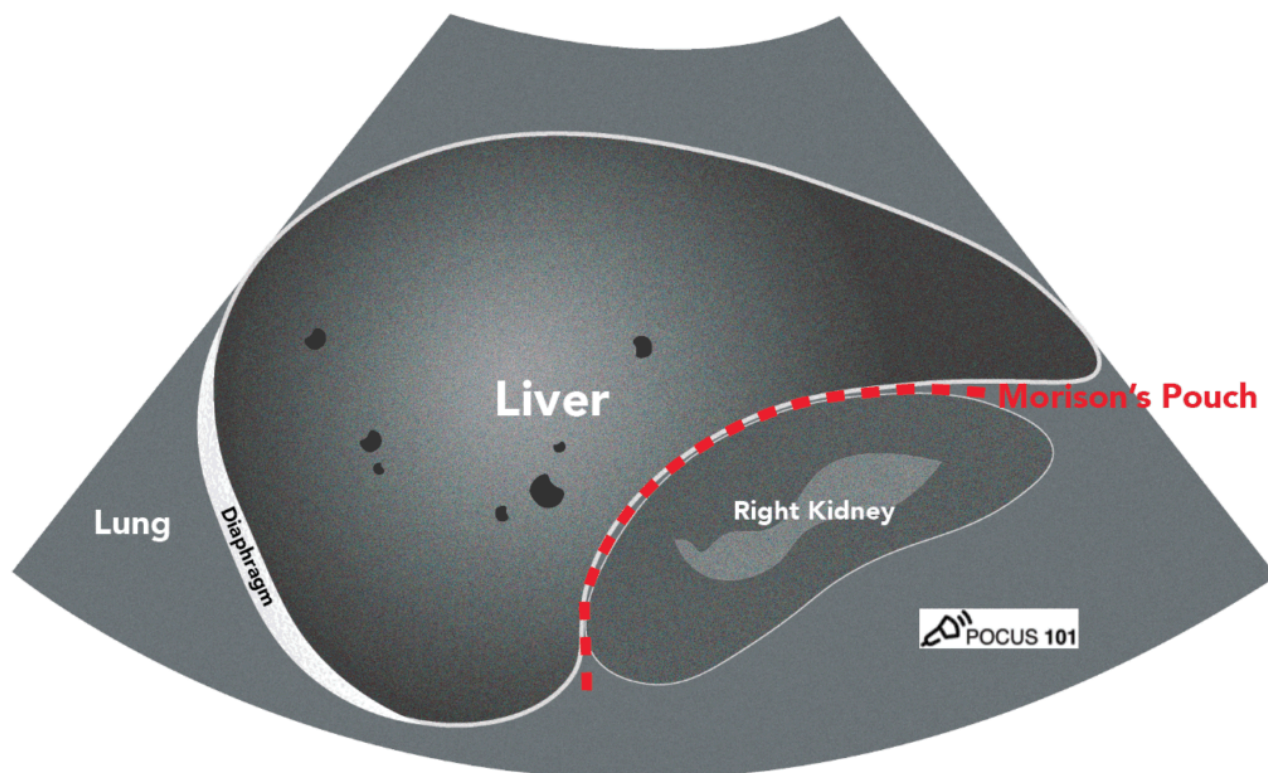
The eFAST is moderately sensitive (approximately 80%) and highly specific (>90%) for detecting free fluid from hemoperitoneum. The general consensus is that there needs to be at least 200-250ml of blood before the eFAST scan will appear positive.

Another important point to remember for the eFAST scan is that **observing free fluid on the eFAST scan does not localize the bleeding to a specific organ**. *For example, if free fluid is noted in the pelvis, it could be originating from anywhere in the abdomen, and **does not** localize the injury to the bladder.* A CT scan is needed to localize the origin of abdominal bleeding in a trauma patient.

Right Upper Quadrant (RUQ) – Hemoperitoneum

The three common locations for free fluid to accumulate in the RUQ of the eFAST scan are the:

- Hepatorenal Space or “Morison’s Pouch”
- Caudal Tip of the Liver
- Suprahepatic Space



eFAST RUQ Morison's Pouch



Free Fluid at the Caudal Tip of the Liver



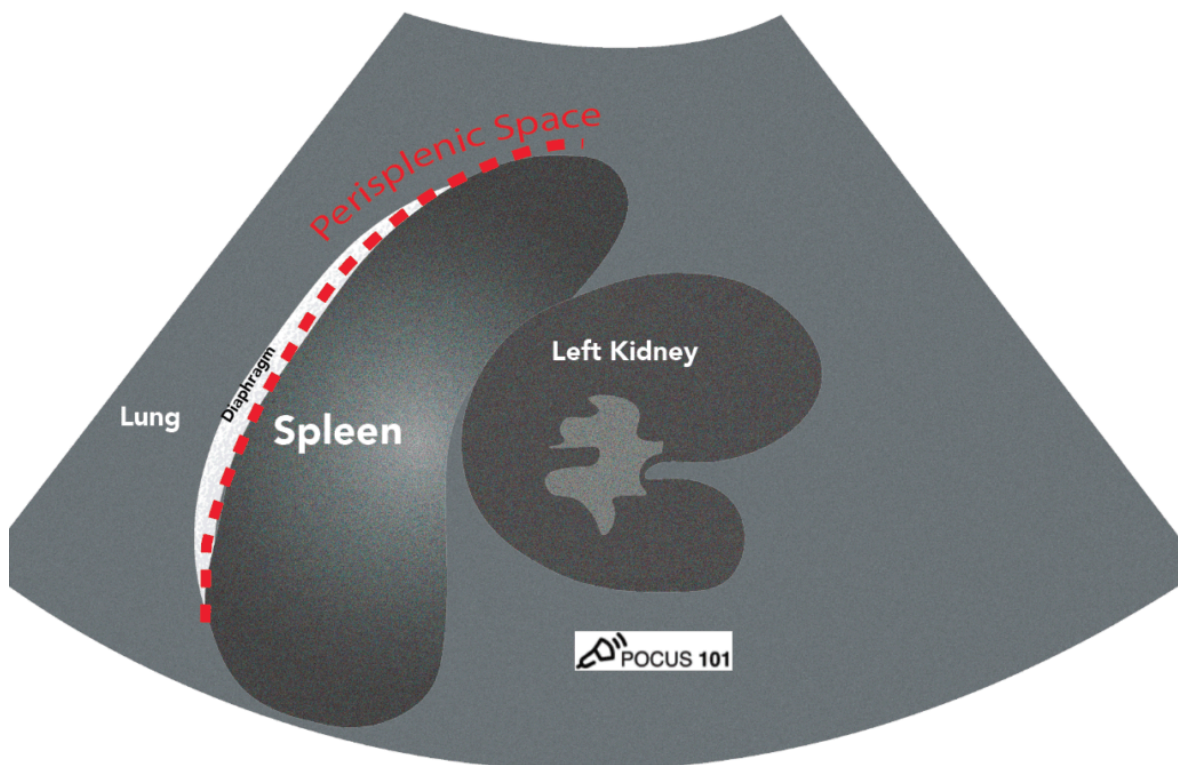
Free Fluid in Morrison's Pouch and Suprahepatic Space

Left Upper Quadrant (LUQ) – Hemoperitoneum

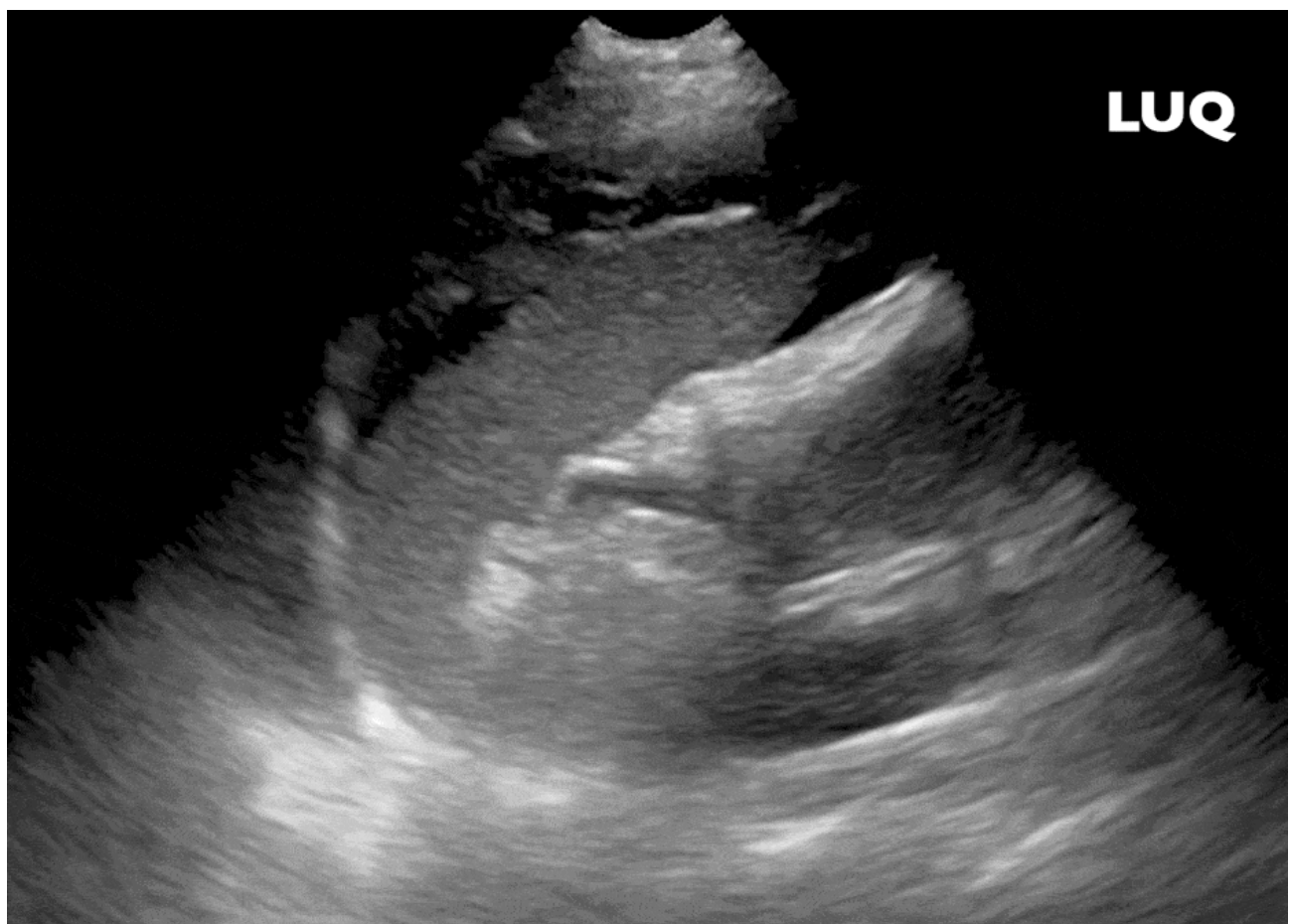
We will evaluate the LUQ in the eFAST for free fluid in the following places:

- **Perisplenic Space**
- **Spleen Tip**
- **Splenorenal Recess**

POCUS 101 TIP: It is important to note that in the LUQ the most common area to find fluid is in the perisplenic space, NOT between the spleen and the left kidney. This is because there is a splenorenal ligament that attaches the spleen and the left kidney preventing a significant amount of fluid to accumulate there unless the ligament is ruptured.



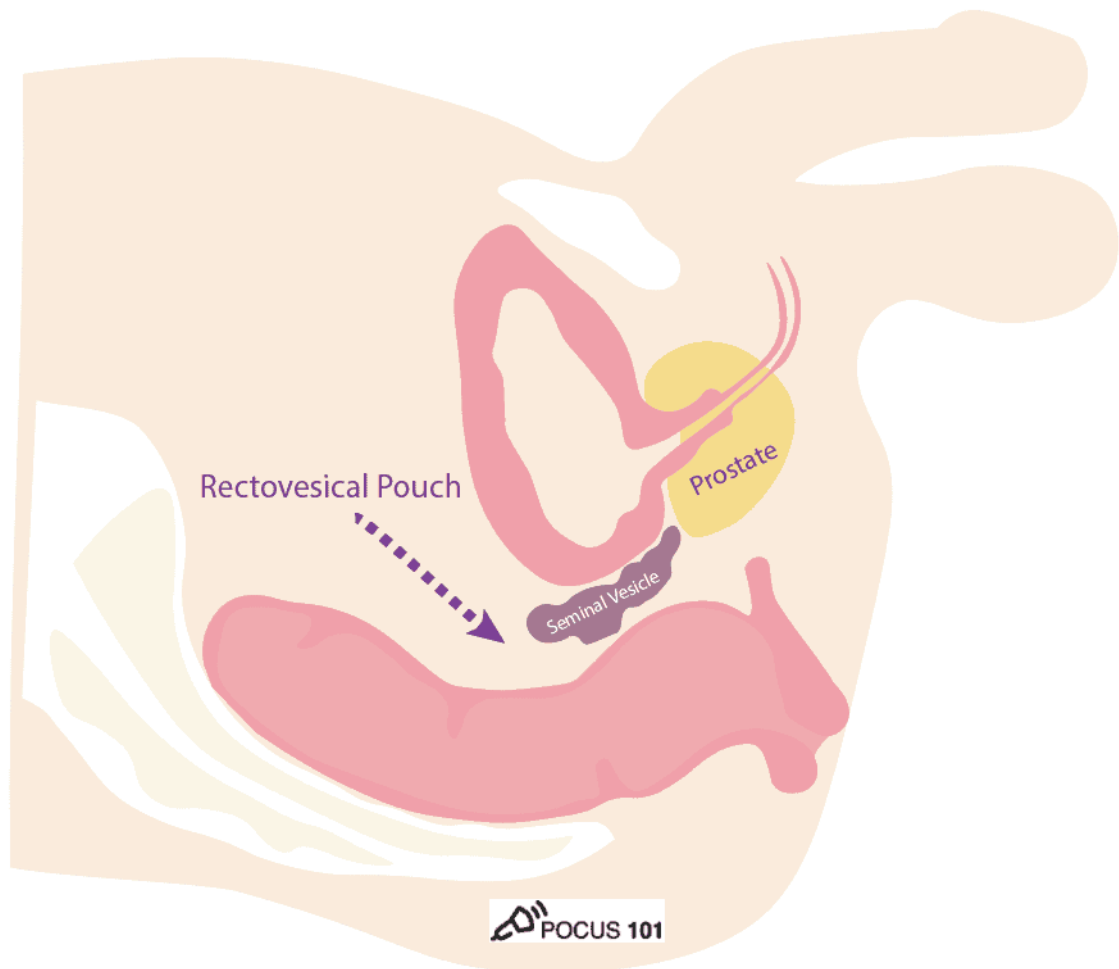
eFAST LUQ Perisplenic Space



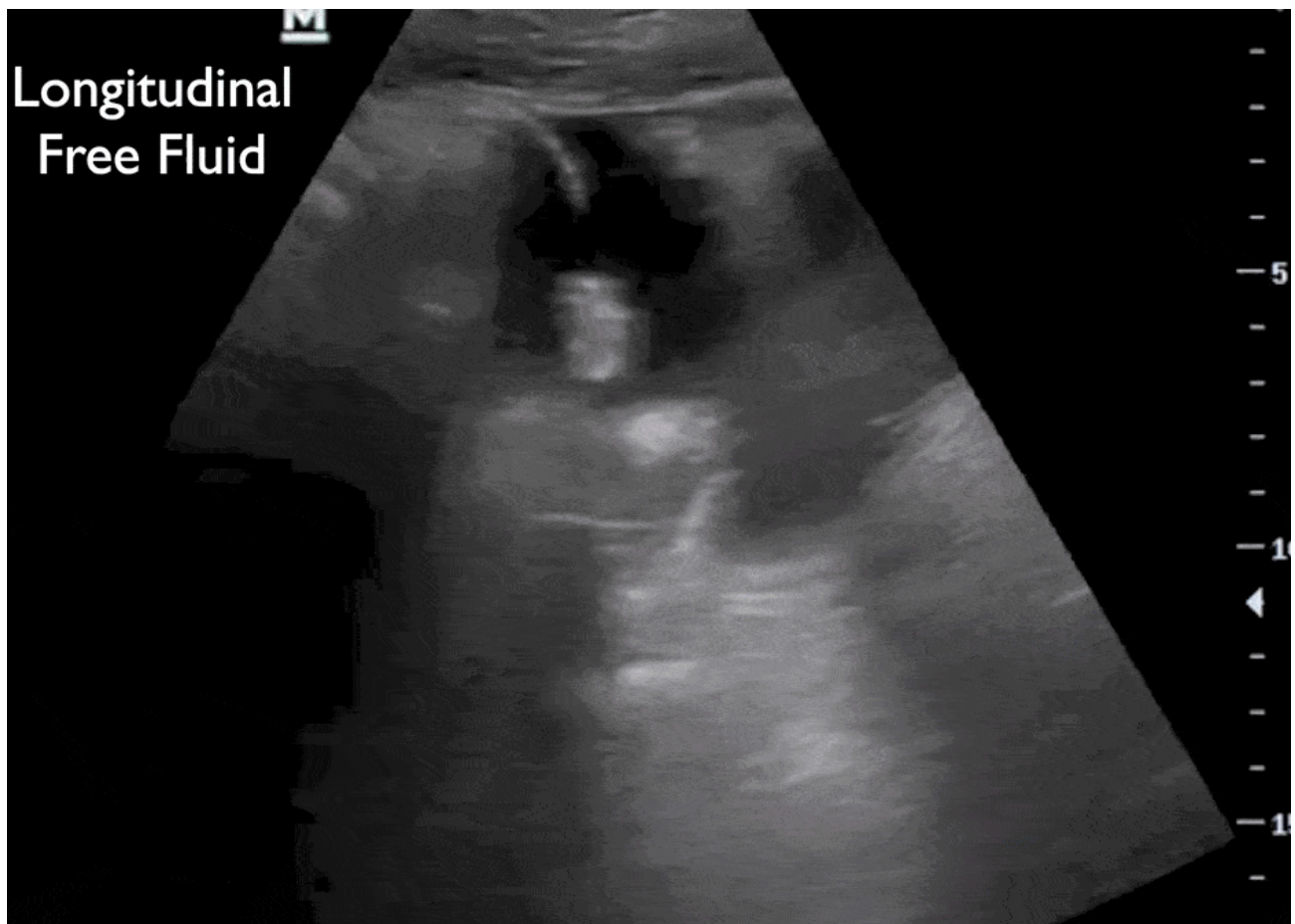
Free fluid in Perisplenic Space

Male Pelvis – Hemoperitoneum

In the male pelvis, you can find free fluid in the rectovesical pouch/space.



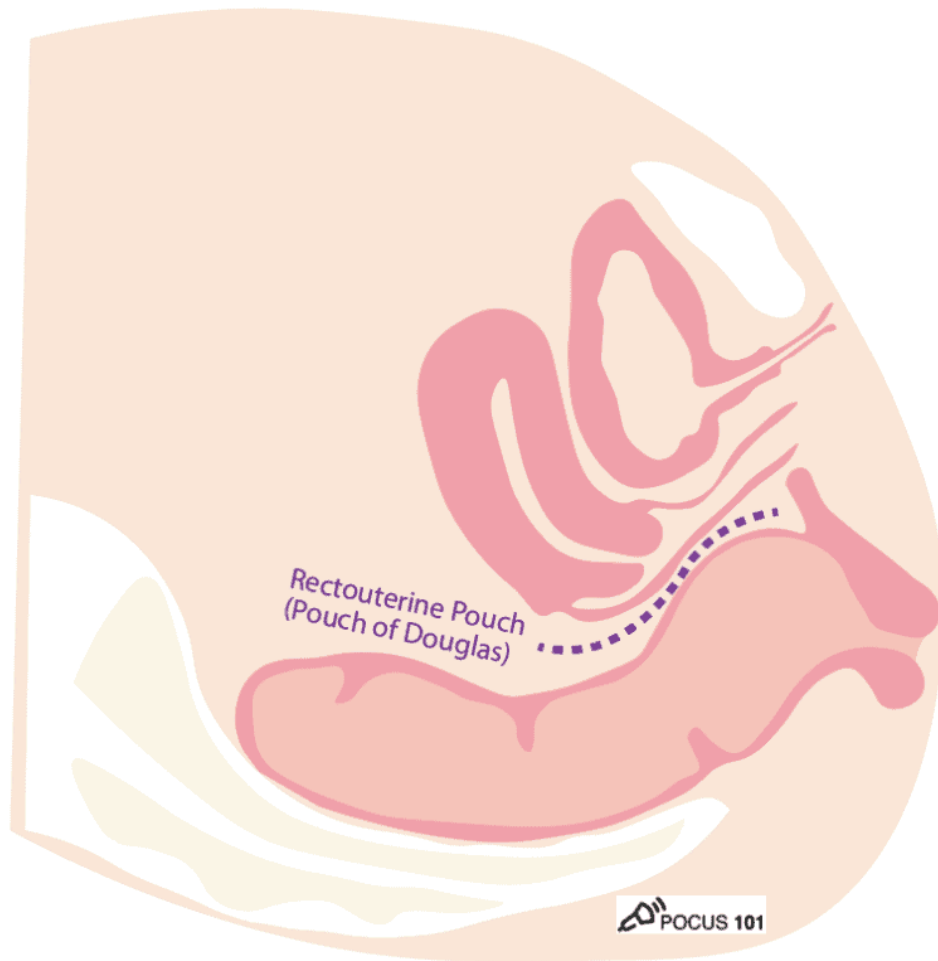
Male Pelvis Rectovesical Pouch



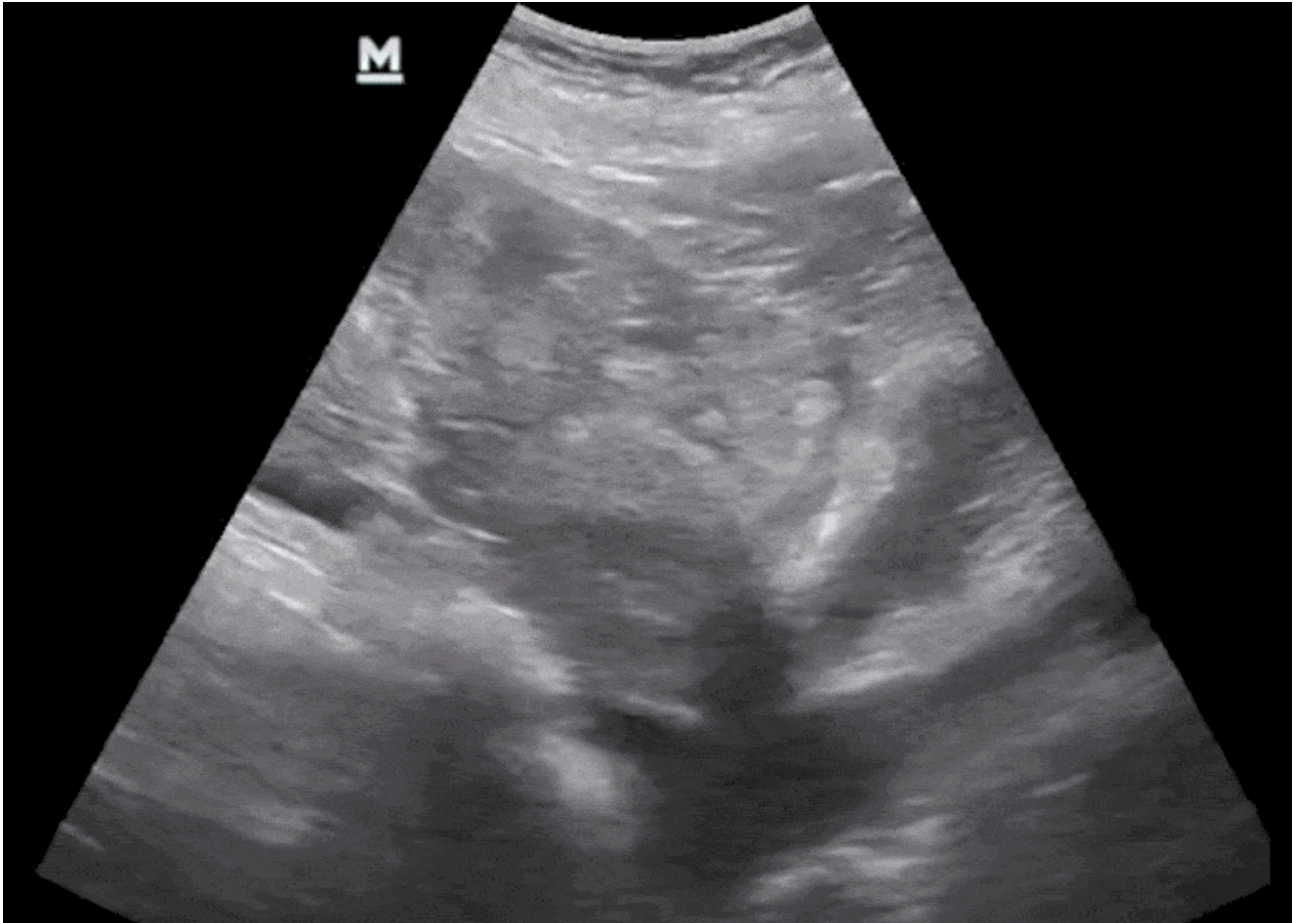
Male – Abdominal Free Fluid in Pelvis (Rectovesical Pouch)

Female Pelvis – Hemoperitoneum

In the female pelvis, you can find free fluid in the Pouch of Douglas (Rectouterine Pouch).



Female Pelvis Rectovesical Pouch



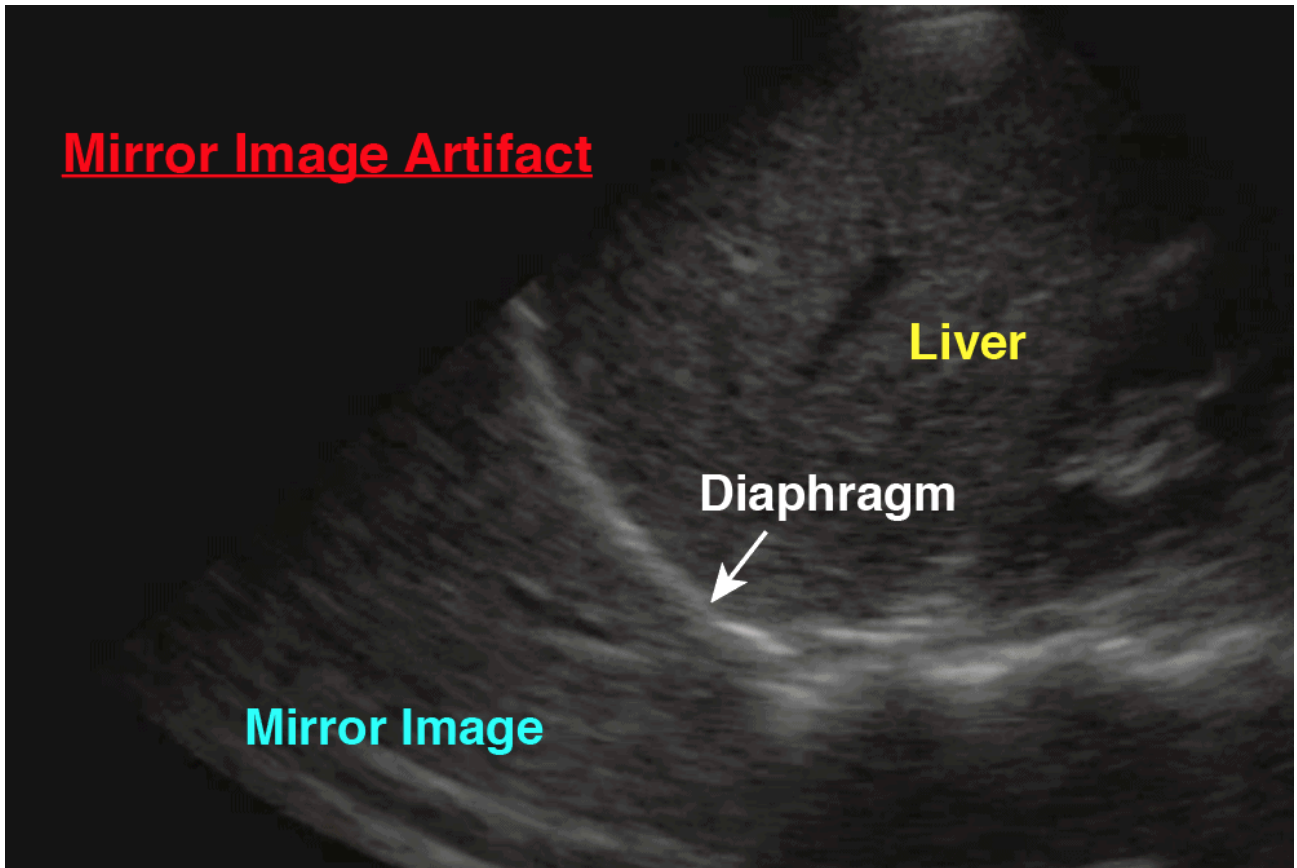
Female – Abdominal Free Fluid in Pelvis (Pouch of Douglas)

Hemothorax– eFAST

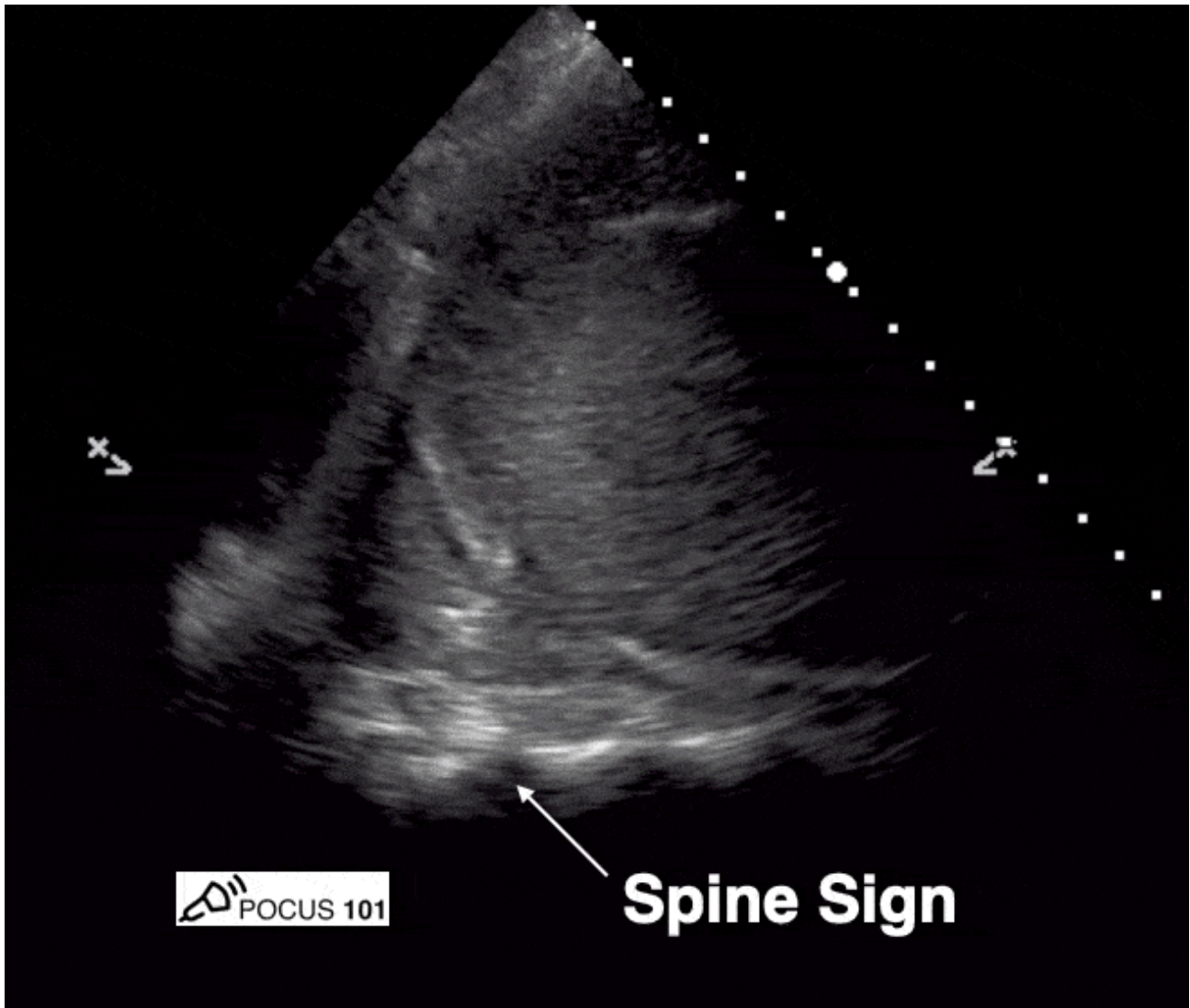
After evaluating the RUQ or LUQ, move the probe superiorly one or two rib spaces to evaluate the thorax for fluid accumulation.

A normal lung will have a Mirror Image Artifact and you will be unable to see the spine going above the diaphragm since all of the ultrasound waves will be reflected back by the aerated lung.

Visualizing the patient's spine **above the diaphragm** implies that there is **free fluid (e.g. blood) in the thorax** since ultrasound waves can easily pass through the free fluid in the chest cavity, allowing you to see the spine. This is referred to as a **Positive Spine Sign** ([click here](#) for a more in-depth explanation on the spine sign).



Absence of Spine. **Normal Finding.**



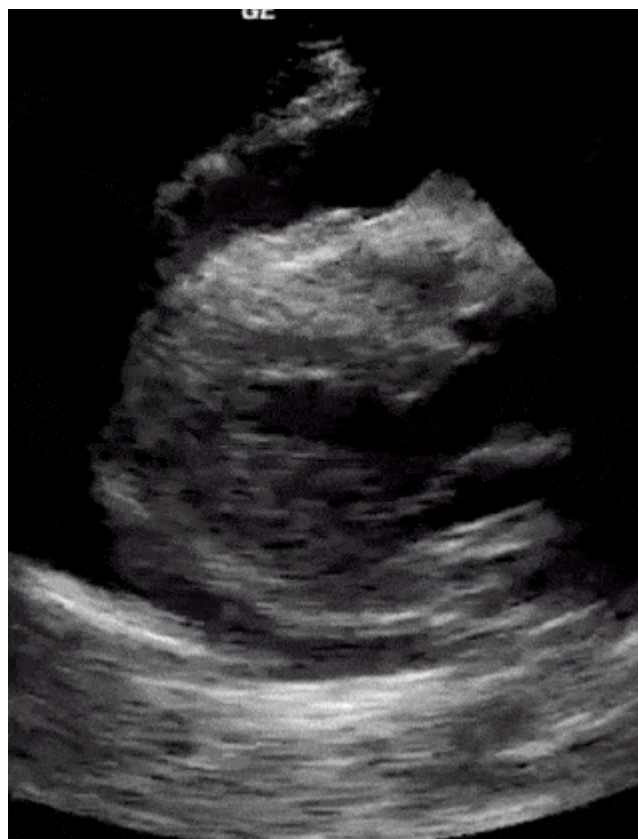
Presence of Spine. **Pathologic Finding.**

Pericardial Effusion and Tamponade – eFAST

- **Anechoic** free fluid can accumulate in the pericardial sac causing a **pericardial effusion**.
- Simply seeing a pericardial effusion not mean the patient has cardiac tamponade. Rather, the **fluid must be impairing cardiac filling for it to be considered tamponade**.
- Consider tamponade when the following is observed:
 - Right Atrial Systolic Collapse** – the **most sensitive** (*and earliest*) echocardiographic finding of tamponade (Perez-Casares, A., et al). Also referred to as the **Trampoline Sign**.
 - Right Ventricle Diastolic Collapse** – the most specific echocardiographic finding for tamponade (75-90%) (Armstrong, et al)



eFAST Subxiphoid View with Pericardial Effusion



eFAST Parasternal Long Axis view with Pericardial Effusion and Tamponade (RV Diastolic Collapse)

Pneumothorax – eFAST

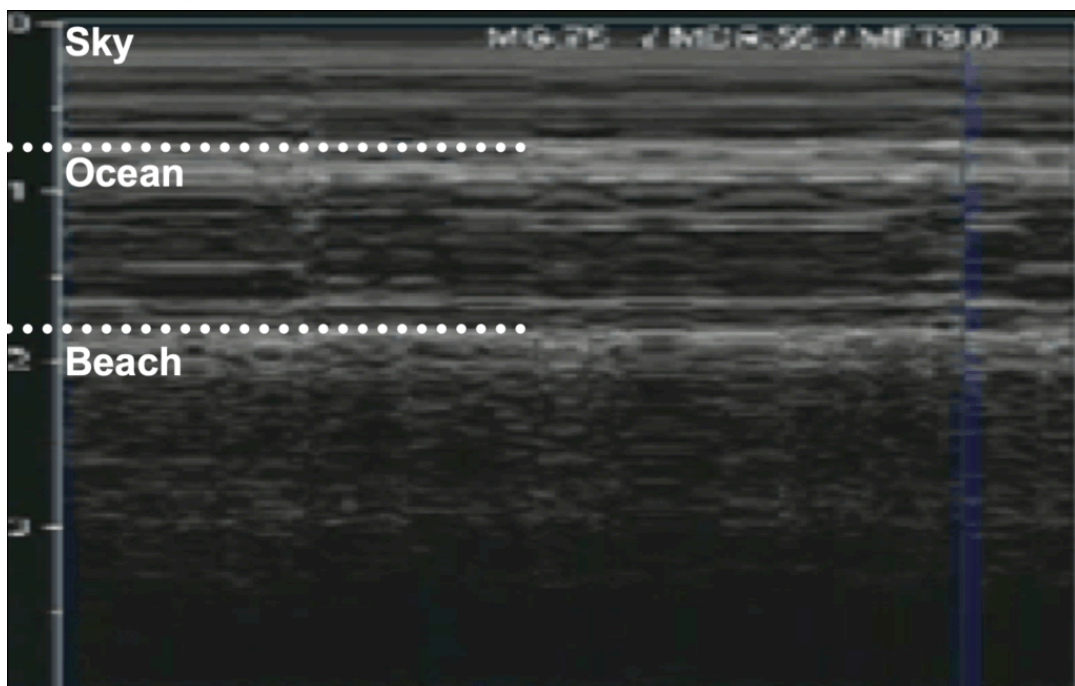
Here are three important steps to evaluating for pneumothorax when performing the eFAST scan:

First, if lung sliding is present, you can rule *out* pneumothorax with 100% accuracy at that ultrasound point (Husain LF).

You can look for lung sliding with B-mode or M-mode:



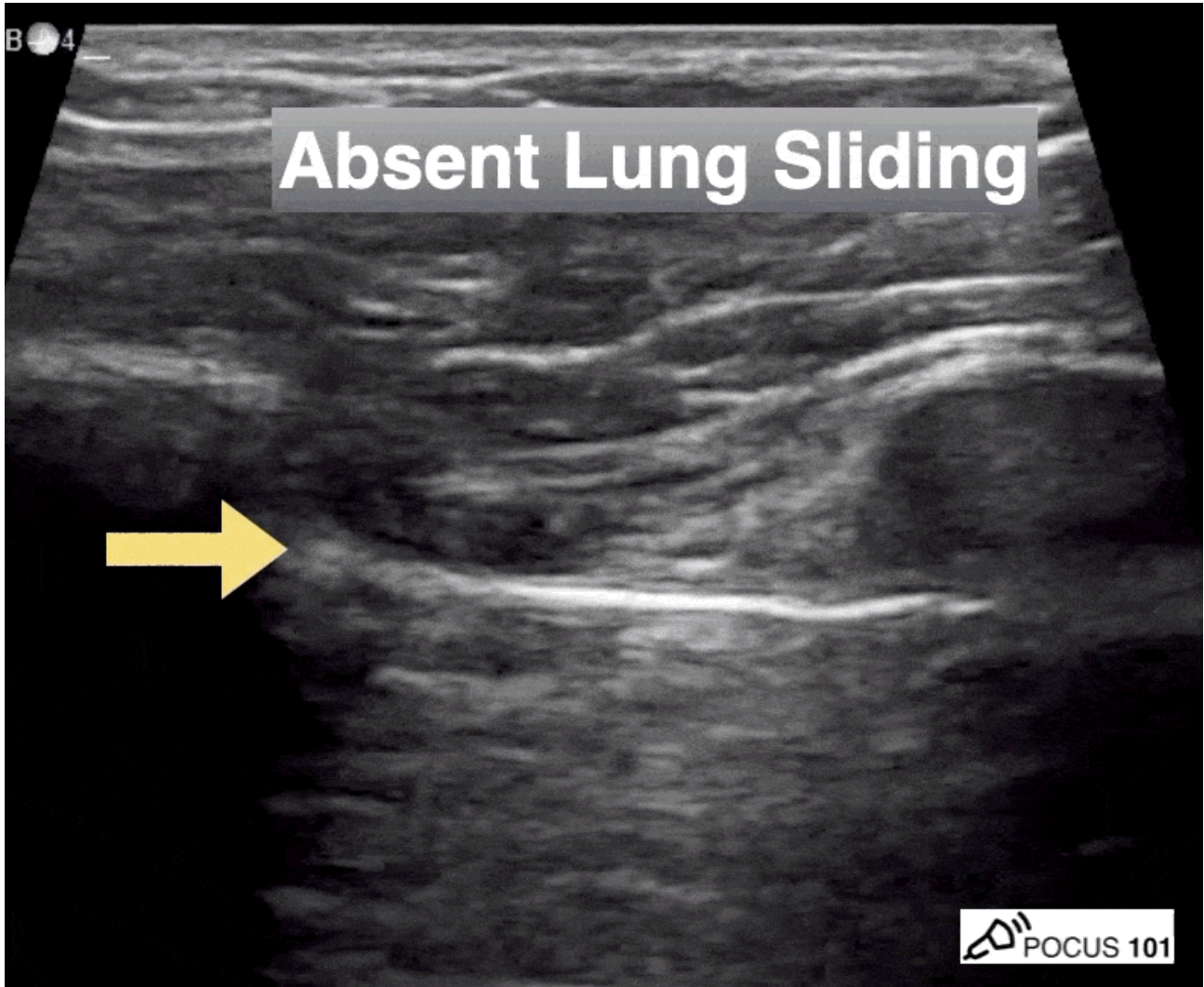
Normal Lung Sliding (B-mode)



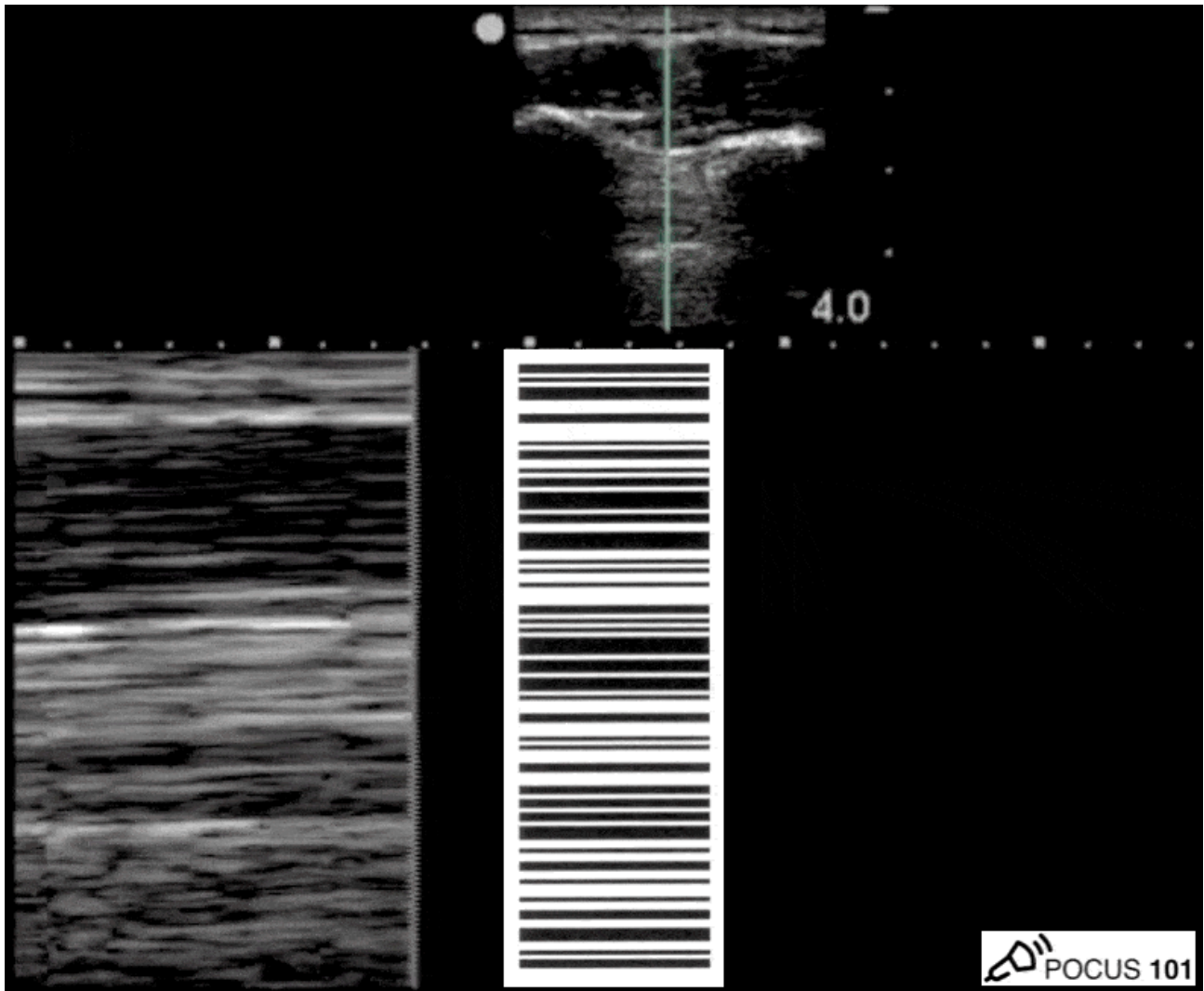
Normal Lung Sliding with Seashore sign (M-mode)

Second, if lung sliding is ABSENT, you should *not automatically assume* pneumothorax.

Recall other causes of reduced/absent lung sliding: severe consolidation, chemical pleurodesis, acute infectious or inflammatory states, fibrotic lung diseases, acute respiratory distress syndrome, or mainstem intubation.



Absence of Lung Sliding (B-mode)



Absence of Lung Sliding – **Barcode Sign** (M-Mode)

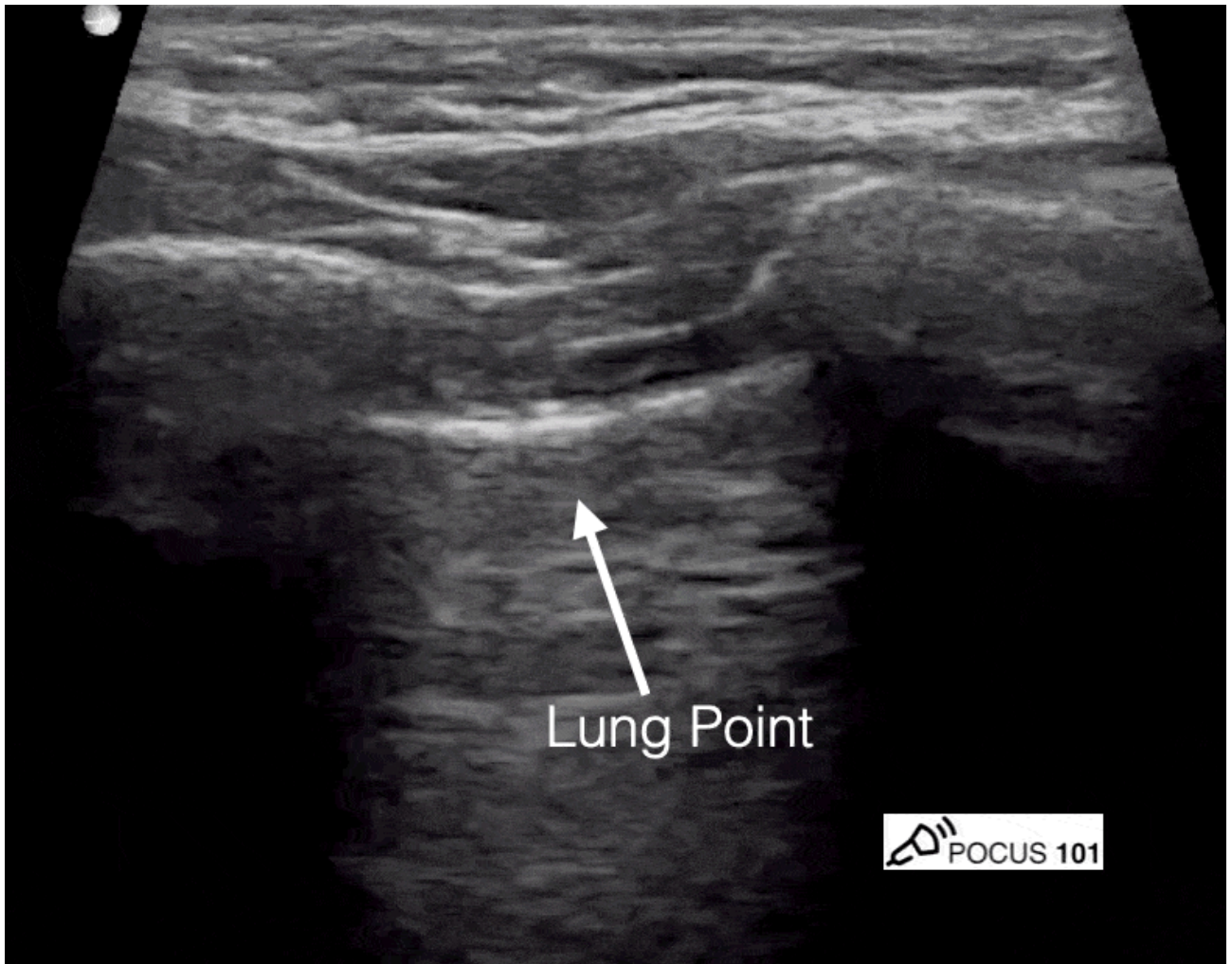
For the purposes of the eFAST scan, it is highly likely that your patient has a pneumothorax if you do not see lung sliding on B-mode or M-mode. If you want to confirm you can proceed to look for the “Lung Point Sign” below.

Third, if a lung point is present, you can rule *in* pneumothorax with 100% accuracy (Chan S).

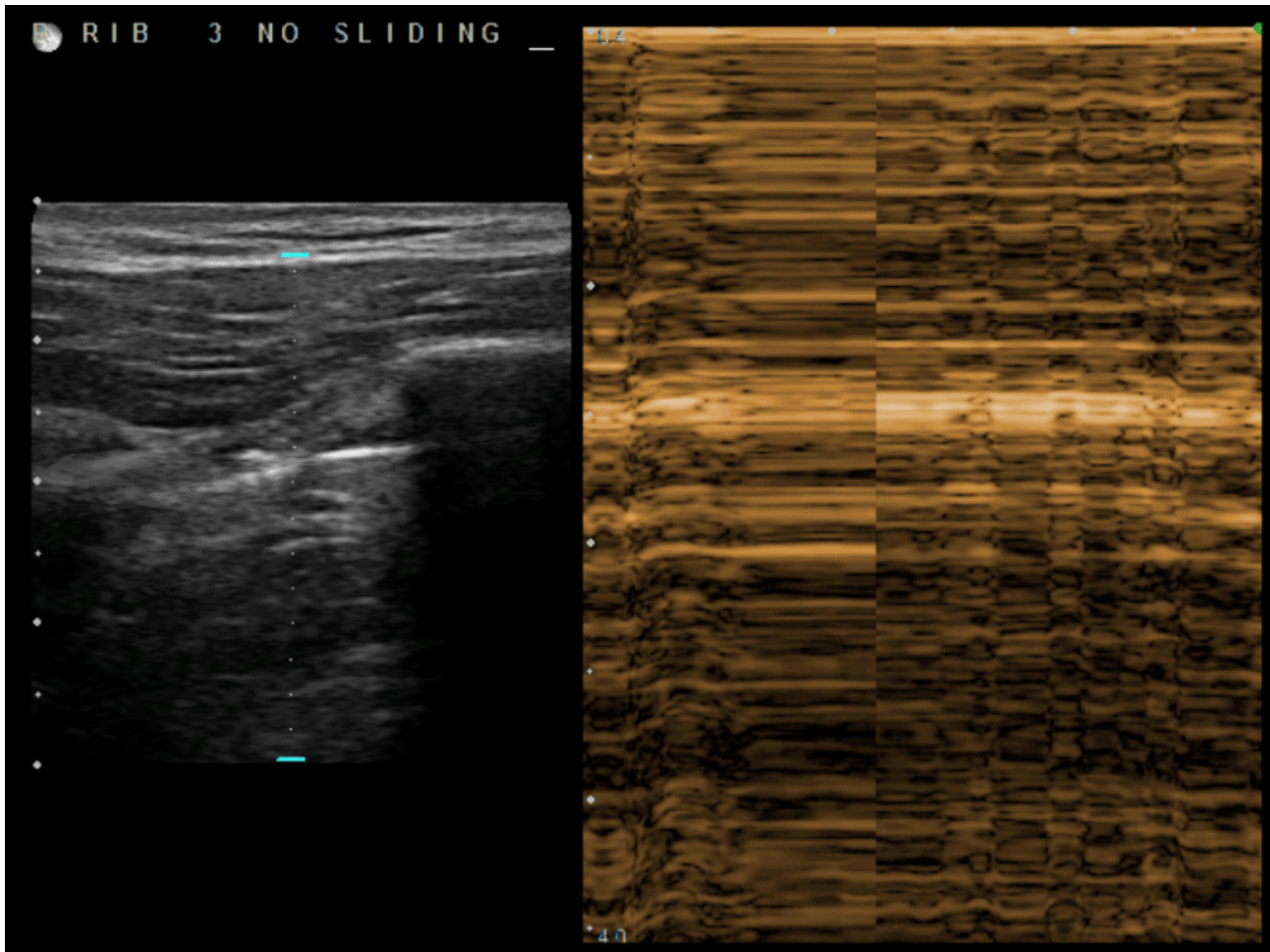
To confirm the presence of a pneumothorax, you should look for the “**Lung Point Sign.**”

The **lung point** is when you can see the transition between normal lung sliding and the absence of lung sliding. This is the transition point between the collapsed lung and normal lung. If you see this you can definitively *rule in* a pneumothorax. The Lung point sign also helps you quantify how large a pneumothorax is.

If you think you may have found a lung point but are not sure, use M-Mode and place your cursor at the intersection where you think lung sliding starts and stops. If you see a normal seashore sign that turns into an abnormal barcode sign, then you have located the lung point with M-Mode.



Lung Point Sign (B-mode)



Lung Point (M-mode)

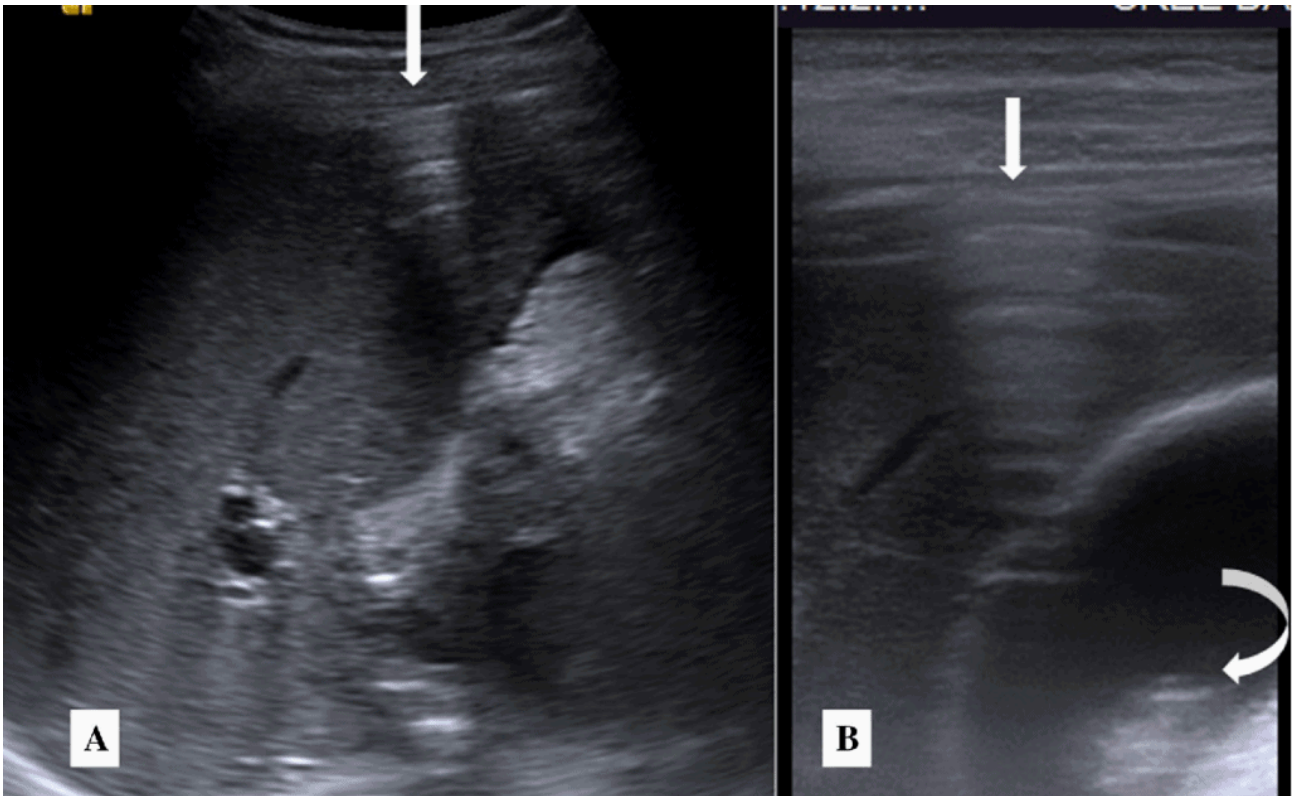
Pneumoperitoneum– eFAST

In the setting of trauma (especially penetrating trauma) you may encounter pneumoperitoneum, or free air within the peritoneal cavity.

On abdominal ultrasound, the most common finding for pneumoperitoneum is the **Enhanced Peritoneal Stripe Sign (EPSS)**. This is when air within the peritoneal space rises and causes an “echoing” of the usually single, hyperechoic peritoneal stripe that separates the abdominal wall from underlying peritoneal fluid and fluid-filled organs (Indiran).

If there is a large amount of pneumoperitoneum, your image of abdominal organs will be obscured by air wherever you place your probe.

POCUS 101 tip: if you can't get any good abdominal views despite having your probe in the correct position, have a high suspicion for pneumoperitoneum.



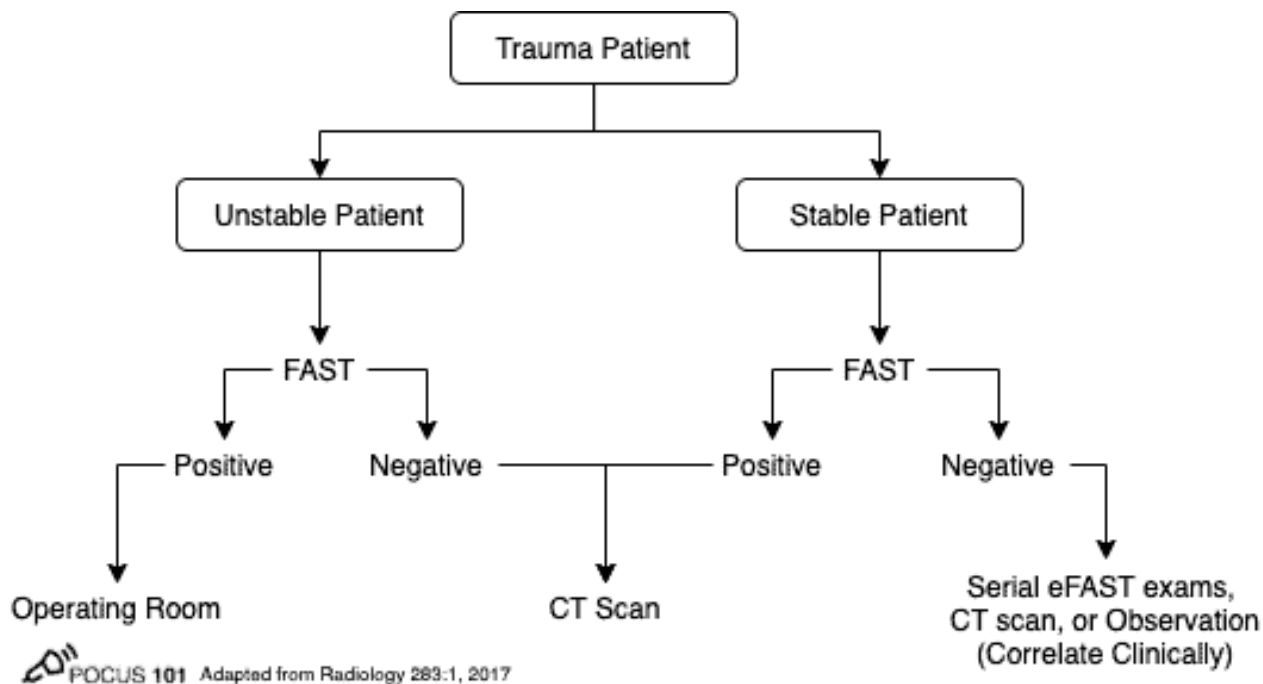
Enhanced peritoneal stripe sign (EPSS) seen anterior to the liver in both images (straight arrows), indicating the abnormal presence of air between the liver and the anterior abdominal wall (Indiran).

eFAST Algorithm and Summary

- Remember that the POCUS eFAST ultrasound exam is most beneficial in hemodynamically unstable patients who are unable to go to the CT scanner. A positive eFAST scan can help the surgeon identify the general region of bleeding (i.e. abdomen vs heart vs lungs) to plan their surgical approach.
- A **negative initial eFAST** exam in patients with a *highly-suspicious mechanism of injury may benefit from a CT scan or serial eFAST exams*, especially in the context of a worsening clinical status (e.g. worsening vitals, hemodynamic instability, worsening pain, or worsening abdominal exam) as patients can also have a delayed presentation.

eFAST Exam Algorithm

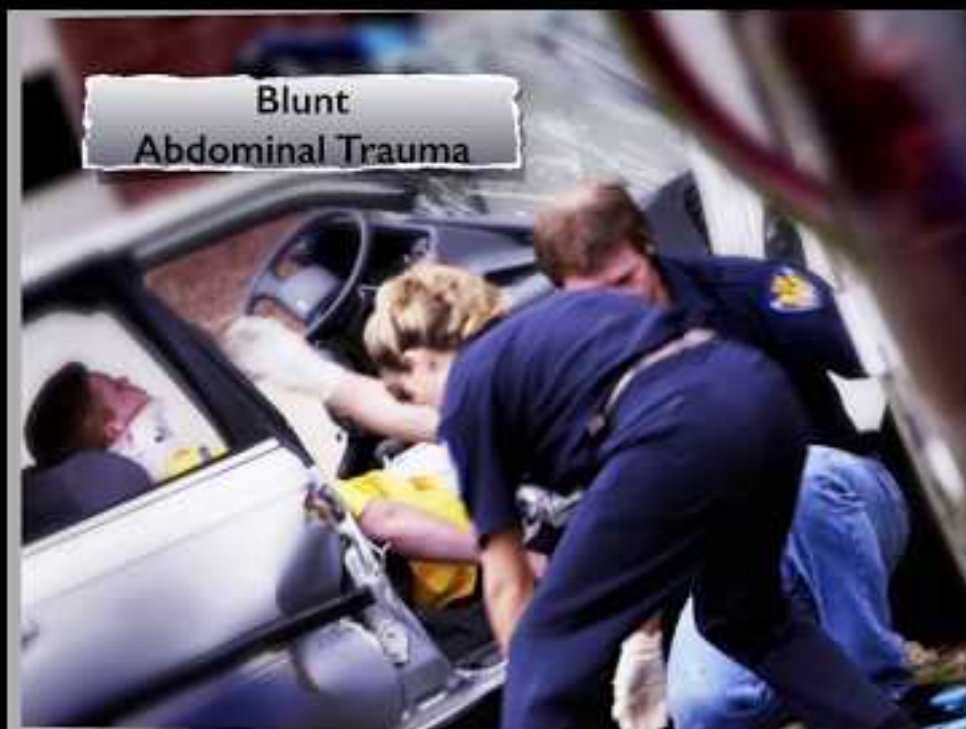
Here is an algorithm for how to use the eFAST exam in the clinical setting:



eFAST Scan Algorithm

Summary Video of eFAST Ultrasound Exam

We made this video to completely summarize how to perform and interpret the eFAST Ultrasound Scan:



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